

# Automorphisms of the Surcomplex Numbers

Real forms, Hahn symmetries,  
valuation, and rigidity

Research article prepared with ChatGPT

22 September 2026

Built from one manuscript of batch 20 of the collection, pinned to repository commit dcf8666

## Abstract

The phrase “automorphism of the surcomplex numbers” has several inequivalent meanings. For the algebraically closed field  $\mathbb{K} = \mathbf{No}(i)$ , the automorphisms fixing  $\mathbf{No}$  pointwise are just identity and conjugation. Those preserving  $\mathbf{No}$  setwise form  $\text{Aut}(\mathbf{No}) \times C_2$ , but this is a proper subgroup of the unrestricted field automorphisms. We prove this distinction explicitly, using Hahn-series phase twists that fix every ordinary complex number and every valuation value while moving the surreal real axis.

We develop the coefficient, character, value-group, and leading-term-kernel layers of the valued automorphism group; construct automorphisms moving ordinary real constants by infinitesimals; and separate strong summation from fine continuity. A further contrast is that nonidentity surcomplex field automorphisms can be everywhere fine-differentiable with derivative zero. We examine class-sized homogeneity, fixed fields, involutions, inequivalent real forms, norm and circle preservation, simplicity, the omega-map, and exponential structure. The repository’s existing exponential faithfulness argument is incorporated as prior work, not presented as a new discovery. That repository report classifies the automorphisms preserving the logarithmic modulus,  $\text{Aut}_L(\mathbb{K}) \simeq \text{Aut}_{\text{exp}}(\mathbf{No}) \times C_2$ ; the present report studies plain field automorphisms, of which those form a proper subgroup of the real-axis stabilizer. The final sections identify precise formalization targets and distinguish proved statements, imported theorems, and unresolved classification problems.

**Scope and status.** An AI-assisted research draft, not refereed. A mathematical research exposition, not a claim of complete classification or of historical priority. Written proofs are supplied for the main reductions and constructions. No new Lean verification is supplied here. Its labels have no dedicated implementation mappings; existing generic Lean results already cover the ordered displacement and value-fixing exponential rigidity implications (Section 1.5). The original repository comparison is pinned to the revision recorded in Section 13; the relations to the collection as it now stands are in Section 1.5 and Appendix C. The word “automorphism” is always qualified by the structure it preserves (Section 1.4). Every label carries the prefix `saut`..

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# 1 Which structure is being preserved?

Let  $\mathbf{No}$  be Conway's real closed ordered field of surreal numbers and put

$$\mathbb{K} = \mathbf{No}(i) = \{x + iy : x, y \in \mathbf{No}\}, \quad i^2 = -1.$$

This is the surcomplex field. Its algebraic closedness follows from real closedness of  $\mathbf{No}$ . Write

$$c(x + iy) = x - iy, \quad N(x + iy) = x^2 + y^2, \quad |x + iy| = \sqrt{x^2 + y^2}.$$

The square root is the nonnegative surreal square root. In particular, the modulus is *surreal-valued*, not an ordinary real-valued norm. The underlying surreal construction and its normal forms are classical [1, 2].

The elementary equation  $\alpha(i)^2 = -1$  forces any field automorphism  $\alpha$  of  $\mathbb{K}$  to send  $i$  to  $i$  or  $-i$ . It does *not* force  $\alpha(\mathbf{No}) = \mathbf{No}$ . This is the first logical distinction on which the subject turns. Once  $\mathbf{No}$  is preserved, order is automatic on that subfield; without the distinguished subfield, the pure algebraically closed field has no intrinsic real axis.

Three groups should therefore never be conflated:

$$\text{Aut}(\mathbb{K}/\mathbf{No}) \subsetneq G_{\mathbf{No}} := \{\alpha : \alpha(\mathbf{No}) = \mathbf{No}\} \subsetneq \text{Aut}(\mathbb{K}). \quad (1.1)$$

The left-hand group fixes  $\mathbf{No}$  *pointwise*. The middle group preserves it *setwise*. The right-hand group has no such condition. We will prove

$$\text{Aut}(\mathbb{K}/\mathbf{No}) = \{\text{id}, c\}, \quad G_{\mathbf{No}} \simeq \text{Aut}(\mathbf{No}) \times C_2,$$

and exhibit witnesses to both strict inclusions.

Other reasonable structures produce other answers. One may mark ordinary  $\mathbb{C}$ , the natural valuation ring, every individual valuation value, the Conway monomials, strong Hahn summation, conjugation, modulus, simplicity, a derivation, or a real exponential. Naming one does not silently name all the others. For example, preserving every ordinary complex number and every valuation value still permits a rotation of some real monomials into imaginary ones.

## 1.1 Literature and the repository

Kuhlmann–Serra developed decompositions of valued Hahn-field automorphisms [3]; their work on Hahn groups supplies a second layer of structure [4]. Kaplan–Krapp–Serra apply these ideas to  $\mathbf{No}$ , including strong summation and additional exponential structure [5]. We use these as the structural background, while proving the complex-coefficient reductions and explicit examples below directly.

The supplied repository already contains a manuscript proving faithful value-group action for surreal exponential automorphisms, with a logarithmic-modulus application to  $\mathbf{No}(i)$  [10]. That argument is reproduced with attribution in Section 11. Its result is important: one must not simply repeat the corresponding question in an earlier source as unanswered. Section 15 separates that mathematical issue from publication status and priority.

## 1.2 Foundations: a collection of class maps

We work in a class theory such as NBG with global choice. An individual surreal number is a set, but  $\mathbf{No}$ ,  $\mathbb{K}$ , and the graphs of their automorphisms are proper classes. The notation  $\text{Aut}(\mathbb{K})$  denotes a *definable collection of class maps*; it is not itself a class-object in NBG. A group isomorphism formula means a unique factorization

statement about individual automorphisms, with the stated composition law. This convention avoids forming a nonexistent set or class of all class functions. Likewise, a finite subgroup means a faithful action of an ordinary finite group  $G$ , encoded by one class relation in  $G \times \mathbb{K} \times \mathbb{K}$  whose slices are field automorphisms. No set of proper-class graphs is formed; the fixed field is defined by quantifying over the finite index set  $G$ . For a collection of all class automorphisms, a displayed fixed-field identity is instead read pointwise: it asserts that an element is fixed by every individual map exactly when it belongs to the named field. It does not invoke class comprehension with a quantifier over class maps.

Global choice is used for the nonconstructive extension arguments in Section 9. The explicit Hahn maps need no class-sized transcendence basis: their values are defined on the set-sized support of one input at a time. Every infinite sum in this article has a set-sized index family. No proper-class support is admitted.

For a set-based treatment, most local algebra and all explicit formulas can instead be read in a Hahn field  $\mathbb{C}((t^\Gamma))$  for a set-sized divisible ordered abelian group  $\Gamma$ . A self-map requires the chosen parameters to preserve that workspace. Global homogeneity and the topology of the full surreal field should not be transferred to an arbitrary fixed workspace.

### 1.3 Notation

Throughout the full-field discussion,

$$\Gamma = (\mathbf{No}, +, <), \quad t^g := \omega^{-g}.$$

Thus  $t = \omega^{-1}$  is positive infinitesimal. The sign in this convention is essential. In  $t$ -notation, supports increase and are well-ordered; in  $\omega$ -notation, normal-form exponents decrease.

| Symbol                            | Meaning                                                         |
|-----------------------------------|-----------------------------------------------------------------|
| $\text{Aut}(\mathbb{K}/E)$        | Field automorphisms fixing the indicated subfield $E$ pointwise |
| $G_{\mathbf{No}}$                 | Setwise stabilizer of the distinguished surreal real axis       |
| $G_v$                             | Automorphisms stabilizing the natural valuation ring            |
| $v, \text{lc}$                    | Minimum support exponent and its nonzero coefficient            |
| $\mathcal{U}$                     | Automorphisms fixing the leading term of every nonzero input    |
| $\text{Aut}_{\text{ord}}(\Gamma)$ | Ordered <i>additive-group</i> automorphisms of $\Gamma$         |
| $\text{st}$                       | Residue/standard-part map on the finite subring                 |
| $c$                               | The chosen conjugation $x + iy \mapsto x - iy$                  |

### 1.4 Conventions: one meaning per word

This subsection and the next were written when the manuscript joined the collection (Appendix C). They fix vocabulary and state relations. The only mathematics they add is the elementary observation (1.2) below, which is marked as an observation of the edit.

**Kinds of automorphism.** An unqualified “automorphism” of  $\mathbb{K}$  or of  $\mathbf{No}$ , and the notation  $\text{Aut}(\mathbb{K})$ ,  $\text{Aut}(\mathbf{No})$ , always mean a *plain field automorphism*: a bijective ring map, with no further condition. Every other kind is named by the structure it preserves.

- A *valued automorphism* is a field automorphism  $\alpha$  with  $\alpha(\mathcal{O}) = \mathcal{O}$ , that is, an element of the group  $G_v$  of Section 6. Equivalently  $v(\alpha z) = \tau_\alpha(vz)$  for an ordered automorphism  $\tau_\alpha$  of the value group. A valued automorphism need not fix any value. Every field automorphism of  $\mathbf{No}$  is valued (Lemma 2.1 and Section 6); a field automorphism of  $\mathbb{K}$  need not be (Proposition 9.3).

- A valued automorphism is *value-fixing* when  $\tau_\alpha = \text{id}$ , that is,  $v(\alpha z) = v(z)$  for every  $z \neq 0$ . This is the meaning of “exactly value-preserving” and of “values fixed” in Theorem 4.2 and Appendix A.
- A *1-automorphism* is an element of  $\mathcal{U}$  in (5.1) (or of  $\mathcal{U}_{\mathbb{R}}$  for **No**):  $v(\alpha z - z) > v(z)$  for every  $z \neq 0$ , which is the same as fixing every leading term. It is value-fixing, and it fixes  $i$ . The phase twists  $P_\theta$ ,  $\theta \neq 0$ , are value-fixing but are not 1-automorphisms. This leading-term notion is the one of Kaplan–Krapp–Serra [5] that the collection’s rigidity report checked against the pinned version (its Corollary 5.2 uses only that a 1-automorphism fixes every value).
- An *exponential automorphism* is a field automorphism  $\sigma$  of **No** with  $\sigma(\exp x) = \exp(\sigma x)$  for all  $x \in \mathbf{No}$ ; they form  $\text{Aut}_{\text{exp}}(\mathbf{No})$ . The notion is defined for **No** only: no surcomplex exponential is used in this report. By (11.2), the identity is the only exponential automorphism that is value-fixing, and a fortiori the only exponential 1-automorphism.
- An *L-automorphism* is a field automorphism  $\alpha$  of  $\mathbb{K}$  with  $\alpha(L(z)) = L(\alpha z)$  for every  $z \neq 0$ , where  $L(z) = \log |z|$  (11.3). They form  $\text{Aut}_L(\mathbb{K})$ , the notation of the rigidity report; the source wrote  $\text{Aut}(\mathbb{K}, L)$ .
- *Strong* and *strongly  $\mathbb{C}$ -linear* are the support conditions of Section 3. They are not continuity conditions (Section 7).

The inclusions that hold are

$$\mathcal{U} \subseteq \{\text{value-fixing}\} \subseteq G_v \subseteq \text{Aut}(\mathbb{K}), \quad \{\text{id}, c\} = \text{Aut}(\mathbb{K}/\mathbf{No}) \subseteq \text{Aut}_L(\mathbb{K}) \subseteq G_{\mathbf{No}} \subseteq G_v.$$

The first chain is immediate from the definitions. In the second, the middle inclusion is the range argument of Theorem 11.4 and the last is Proposition 7.1. *Observation of the edit.* Combining Theorem 2.2 with Theorem 11.4,

$$\text{Aut}_L(\mathbb{K}) = \{\alpha \in G_{\mathbf{No}} : \alpha|_{\mathbf{No}} \in \text{Aut}_{\text{exp}}(\mathbf{No})\} \subsetneq G_{\mathbf{No}}. \quad (1.2)$$

The equality is Theorem 11.4 read through the factorization (2.3). Strictness is witnessed by  $S_q$  for any rational  $q > 0$ ,  $q \neq 1$ : it lies in  $G_{\mathbf{No}}$  (Section 4), and its restriction to **No** induces the value-group dilation  $g \mapsto qg$ , which no exponential automorphism induces (Proposition 11.3). Whether the first inclusion  $\{\text{id}, c\} \subseteq \text{Aut}_L(\mathbb{K})$  is strict is the question whether  $\text{Aut}_{\text{exp}}(\mathbf{No})$  is nontrivial; neither this report nor the rigidity report decides it.

**Rigidity.** In this report the word labels three specific statements: exact norm rigidity (Theorem 2.3), the simplicity statement of Section 11, and the faithful value-group action of exponential automorphisms (Theorem 11.2); the last is faithfulness, not triviality, exactly as in the rigidity report. None of them is the all-scale polynomial rigidity `f:thm-rigidity of surcomplex/analysis` or the scalar rigidity of `surreal/gamma-functions`.

**Phase.** The phases of the twists  $P_\theta$  are the coefficient multipliers  $e^{i\theta\ell(g)}$  of (4.6). They are not an argument of  $z$ , not the finite-angle `cis` or the global phase of `surcomplex/trigonometry`, and not the phase of the differential-equations report; `NOTATION.md` lists those meanings.

**Real form.** A real form is a real closed subfield  $F \subset \mathbb{K}$  with  $\mathbb{K} = F(i)$  (Section 10). In `ent:prop:real-form of surcomplex/entire-functions-at-arbitrary-rank` and in `analytic:subsec:fixed-real of surcomplex/analytic-geometry` the phrase names the real-coefficient version of a theorem, an unrelated use.

**Symbols.** Here  $\Gamma = (\mathbf{No}, +, <)$  is a proper class; this departs from the recommendation of `NOTATION.md` that  $\Gamma$  be a set-sized group, and the set-sized reading is the one described in the previous subsection. The conventions  $t^g = \omega^{-g}$ ,  $v = \min \text{supp}$  and  $v(0) = \infty$  are the collection's.  $\mathbb{K} = \mathbf{No}(i)$  is the field that `NOTATION.md` writes  $\text{SC} = \mathbf{No}[i]$  and the analysis report writes  $\mathbb{K}$ ;  $K_1$ ,  $K_2$  and  $F(i)$  in Sections 2 and 9 are generic. Six symbols of the source were renamed so that each has one meaning.

- The source wrote  $E_{\rho, \tau}$  for the coefficient-and-exponent lift and  $E_d$  for the Taylor motion of Theorem 5.1. Here they are  $M_{\rho, \tau} = M_{\rho, \tau, 1}$  and  $\Phi_d$ , so that  $E$  denotes only an ordered exponential (Theorem 11.2), as in the rigidity report.
- The derivation of Theorem 5.2 is  $\mathcal{D}$  (source:  $D$ ). A  $D$  with a subscript is always a diagonal automorphism  $D_\lambda$  or  $D_\chi$ ; the unsubscripted  $D(x) = \sigma(x) - x$  in Lemma 11.1 is the displacement, as in the rigidity report.
- The real Puiseux field of Section 7 is  $\mathcal{P}$  (source:  $P$ ), leaving  $P_\theta$  for the phase twists.
- The automorphism in the non-density argument at the end of Section 7 is  $\psi$  (source:  $\tau$ ), since  $\tau$  always denotes an ordered automorphism of the value group.
- The sides of a cut are  $(\mathcal{L}, \mathcal{R})$  (source:  $(L, R)$ ), since  $L$  is the logarithmic modulus.

$T_s$  denotes the flows of Theorem 5.2; the maps  $T_r$  and  $T_{s_0, c}$  of the rigidity report are value-group maps, and for rational  $q > 0$  the dilation  $S_q$  of (4.4), restricted to  $\mathbf{No}$ , is that report's canonical field lift  $\widehat{T}_q$  (`prop:hahn-lift`). The coefficient  $\ell(g) = [g]_0$  of (4.5) is that report's layer coefficient  $[\gamma]_{s_0}$  at  $s_0 = 0$ . The letter  $H$  is local to three places: the kernel subgroup in the proof of Theorem 11.2, the real closure in Theorem 10.2, and the half-plane  $H_F$  of Section 2.

## 1.5 Relation to other reports of the collection

The source compared itself with the collection at its pin; that comparison is kept in Section 13. The relations below were checked against the collection as it stands after this report was placed.

**Exponential automorphism rigidity.** `surreal/exponential-automorphism-rigidity` studies exponential automorphisms of an ordered field with a total exponential and, in its Section 9,  $L$ -automorphisms of  $F(i)$  for real closed  $F$ ; this report studies plain field automorphisms of  $\mathbb{K}$ . The two meet in Section 11, whose results are that report's and are printed here once, with its proofs.

- Lemma 11.1 is its `lem:amplify` (Lemma 3.1), with the same probe.
- Theorem 11.2 is the faithfulness of its `thm:cofinal` and `cor:faithful` (Theorem 4.1, Corollary 4.2), by the route of `lem:bridge` and `lem:bounded`; the identification of the kernel with the finite surreals after it is `prop:finite-log`. That report proves more: cofinal and coinital displacement, coarsenings, a version needing one visible exponential scale, and an order-free detector.
- The display (11.2) and the sentence after it are its `eq:main-no`, `thm:no` and `cor:question54` (Theorem 5.1, Corollary 5.2): every exponential 1-automorphism of  $\mathbf{No}$  is the identity, the negative answer to Question 5.4 of [5], arXiv version 3. The answer is that report's, not this one's, and the reading of the question rests on its check of the pinned PDF.
- Proposition 11.3 is its `thm:dilation` (Theorem 8.4), by the second, multiplication-based of its two proofs, with an arbitrary positive infinite  $X$  in place of  $\omega$ .
- Theorem 11.4 is its `thm:L-classification` with `lem:range` and the kernel clause of `thm:complex-kernel` and `cor:surcomplex` (Lemma 9.1, Theorems 9.2 and 9.4, Corollary 9.5), spe-

cialized to  $F = \mathbf{No}$ . That report proves it for every real closed  $F$  with an ordered exponential, and on every invariant nontrivial convex value quotient.

What this report adds on the surcomplex side is the plain-field picture around that classification. Theorem 2.2 is the field-level counterpart:  $G_{\mathbf{No}} \simeq \text{Aut}(\mathbf{No}) \times C_2$ , with  $\text{Aut}(\mathbf{No})$  in place of  $\text{Aut}_{\text{exp}}(\mathbf{No})$ . That report remarks that a shorter route to its classification, assuming commutation with conjugation and preservation of the exponential on the real axis from the start, proves strictly less; Theorem 2.2 is the exponential-free first step of that shorter route. By (1.2),  $\text{Aut}_L(\mathbb{K})$  is a proper subgroup of  $G_{\mathbf{No}}$ , which in turn is a proper subgroup of  $G_v$  and of  $\text{Aut}(\mathbb{K})$  (9.2). By the kernel clause of Theorem 11.4, an  $L$ -automorphism that fixes every value is  $\text{id}$  or  $c$ . The explicit maps  $D_\lambda$ ,  $P_\theta$ ,  $\Phi_d$  and  $T_s$  are value-fixing and fix  $i$ , so a nonidentity one is never an  $L$ -automorphism; nor is  $S_q$  for rational  $q \neq 1$ , by (1.2). For other  $a$  this report does not decide whether  $S_a|_{\mathbf{No}}$  is exponential. The open question on the image of exponential automorphisms in the value group (Section 14) is that report's `q:image` (Question 13.1).

**Foundations and analysis.** The pair construction of  $\mathbb{K}$  and its algebraic closedness are `found:prop:complex of foundations-and-computation/foundations`. Proposition 7.2 is `found:thm:discrete` there, items (1)–(2), with the absence of isolated points; `FORMALIZATION.md` records Lean proofs of those clauses for the actual surcomplex carrier, with lower-universe smallness. The fine derivative of Section 8 is `found:eq:derivative`. The collection's earlier zero-derivative witnesses are locally constant: the indicator of the finite ring in `found:rem:fourtypes` and in `b:L` of `surcomplex/analysis`, with a Lean proof recorded for the former. The dilation  $S_2$  of Theorem 8.2 is a different kind of witness: it is injective, so it is constant on no open set with more than one point, and the full field has no isolated points.

**Diagonal character twists.** The automorphisms  $D_\chi = M_{\text{id}, \text{id}, \chi}$  are the character twists that the collection already uses on set-sized Hahn fields as tools: `spec:lem:finiteindex` of `surcomplex/spectral-theory`, whose Galois group  $\text{Hom}(\Lambda/\Gamma, \mu_q)$  acts by them; `hnd:lem:supportdegree` of `surcomplex/hidden-negative-hermitian-directions`; and the twists built from `bst:lem:characters` in `surreal/transcendence-over-bounded-support`. There the characters take roots of unity or other values on a set-sized group; here  $\chi$  is an arbitrary homomorphism  $\Gamma = \mathbf{No} \rightarrow \mathbb{C}^\times$ , and the twists are placed in the decomposition of Theorem 6.1.

**Unit circle and rotations.** The Cayley map in the proof of Theorem 2.4,  $u = (x-i)/(x+i)$ , is a reparametrization of `trigonometry:thm:cayley` of `surcomplex/trigonometry` ( $u = C(-1/x)$  for  $x \neq 0$ , while  $x = 0$  gives  $u = -1$ ); that report identifies the unit circle with  $\text{SO}(2, \mathbf{No})$  in Section `trigonometry:sec:circle`. Rotations there, and in `surreal/euclidean-three-space`, placed in the same batch, are  $\mathbf{No}$ -linear isometries, not field automorphisms. The contrast is exact: multiplication by a unit-circle element preserves every modulus, whereas by Theorem 2.3 the only field automorphisms preserving every modulus are  $\text{id}$  and  $c$ .

**Lean interfaces.** Besides the two modules the source inspected (Section 13), the collection has `Surreal/Surcomplex/Basic.lean`, whose `conj` is conjugation as a ring automorphism of the concrete surcomplex field, and `Surreal/HahnSeries/ComplexNumbers.lean`, whose `coefficientEquiv` lifts a ring equivalence of coefficients to Hahn series over an ordered exponent monoid, keeping supports: the case  $M_{\rho, \text{id}, 1}$  of (4.1) in a workspace. `Surreal/Algebra/SigmaDerivation.lean` proves the valued-field forms of the rigidity report's displacement results. The later `Surreal/Algebra/ExponentialProfile.lean` proves its ordered displacement lemma (`displacement_cofinal`) and the value-fixing exponential rigidity implication (`eq_id_of_commute`). These generic results already cover the corresponding finite arguments here; their mappings use the rigidity report's labels. No dedicated `saut: implementation` mapping is added. Instantiation

at the actual surreal exponential, rational non-lifting and the classification for the actual surcomplex logarithmic modulus remain pending; no class automorphism-group object follows merely from a theorem about individual type-based maps.

## 2 Conjugation and the real-axis stabilizer

The results of this section apply to any real closed ordered field  $F$ , with  $K = F(i)$ . No infinite summation or specifically surreal theorem is needed.

**Lemma 2.1.** *Every field automorphism of a real closed field is order preserving. It fixes its real algebraic numbers pointwise.*

*Proof.* In a real closed field,  $x > 0$  is equivalent to  $x = y^2$  for some  $y \neq 0$ . A field automorphism preserves this condition and hence preserves order. It fixes  $\mathbb{Q}$ . Each real algebraic number is isolated from the other real roots of its minimal polynomial by a rational interval, so it is fixed too.  $\square$

**Theorem 2.2** (Centralizer and stabilizer). *For  $K = F(i)$  with  $F$  real closed,*

$$\text{Aut}(K/F) = \{\text{id}, c\}, \quad (2.1)$$

$$\begin{aligned} \{\alpha \in \text{Aut}(K) : \alpha(F) = F\} &= C_{\text{Aut}(K)}(c) = N_{\text{Aut}(K)}(\langle c \rangle) \\ &\simeq \text{Aut}(F) \times C_2. \end{aligned} \quad (2.2)$$

The pair  $(\sigma, e)$ , with  $e \in \{0, 1\}$ , acts by

$$(x + iy) \mapsto \sigma(x) + (-1)^e i \sigma(y). \quad (2.3)$$

*Proof.* The roots of  $X^2 + 1$  in  $K$  are exactly  $i, -i$ . Thus an automorphism fixing  $F$  has precisely the two possibilities in (2.1). If  $\alpha$  preserves  $F$ , its restriction is a field automorphism  $\sigma$  and its value on  $i$  determines (2.3). Conversely, this formula respects addition and the multiplication rule

$$(x + iy)(u + iv) = (xu - yv) + i(xv + yu),$$

and has an inverse of the same form. The lift  $\widehat{\sigma}$  fixing  $i$  commutes with  $c$ , so the product is direct.

If  $\alpha$  commutes with  $c$ , it carries the fixed field  $\text{Fix}(c) = F$  onto itself. The converse follows from (2.3). Finally, normalizing the order-two subgroup  $\{\text{id}, c\}$  means fixing its unique nonidentity member under conjugation, which is the centralizer condition.  $\square$

For surcomplex numbers,  $\text{Aut}(\mathbf{No})$  is very far from trivial. In Section 4, exponent dilation will give a real-axis-preserving automorphism sending  $\omega$  to  $\omega^2$ . Thus even retaining conjugation leaves much more than the two classical-looking maps.

### 2.1 Equivariance is not exact preservation

For  $\alpha = \widehat{\sigma} c^e$  one has

$$N(\alpha z) = \sigma(Nz), \quad |\alpha z| = \sigma(|z|). \quad (2.4)$$

The second identity follows because  $\sigma$  preserves nonnegativity and square roots. In particular, balls are taken to balls, but their radii may change. Calling every such map an isometry would be misleading: it is *semilinearly* compatible with the surreal-valued modulus.

**Theorem 2.3** (Exact norm rigidity). *A field automorphism  $\alpha$  of  $F(i)$  satisfying  $N(\alpha z) = N(z)$  for every  $z$  is either id or  $c$ . The same conclusion holds under exact modulus preservation. No real-axis-preservation hypothesis is needed.*

*Proof.* The identity

$$N(z+1) - N(z) - 1 = 2 \operatorname{Re} z$$

and  $\alpha(z+1) = \alpha z + 1$  imply  $\operatorname{Re}(\alpha z) = \operatorname{Re} z$ . For  $x \in F$ , write  $\alpha x = x + iy$ . Exact norm preservation gives  $x^2 + y^2 = x^2$ , hence  $y = 0$ . Thus  $\alpha$  fixes  $F$  pointwise and [theorem 2.2](#) applies. Exact modulus preservation implies exact norm preservation by squaring.  $\square$

## 2.2 The unit circle remembers the real axis

Let  $U_F = \{z \in F(i) : N(z) = 1\}$ . Merely preserving this set, without preserving individual norms, already recovers the real-axis stabilizer.

**Theorem 2.4.** *For a field automorphism  $\alpha$  of  $F(i)$ ,*

$$\alpha(U_F) = U_F \iff \alpha(F) = F.$$

*Proof.* The reverse implication follows from (2.4). For the forward implication use the Cayley parametrization

$$u = \frac{x-i}{x+i}, \quad x = i \frac{1+u}{1-u}.$$

For every  $x \in F$ , the first expression lies in  $U_F \setminus \{1\}$ . For every  $u \in U_F \setminus \{1\}$ , the second expression is real: conjugating it and using  $\bar{u} = u^{-1}$  leaves it unchanged. Applying  $\alpha$  to the inverse expression changes  $i$  at most to  $-i$ , so  $\alpha x \in F$ . Apply the same reasoning to  $\alpha^{-1}$  to obtain equality of the axes.  $\square$

Let  $H_F = \{z : \operatorname{Im} z > 0\}$  be the chosen upper half-plane. If a field automorphism preserves  $H_F$  setwise, it also preserves  $-H_F$ , so it preserves their complement  $F$ . Since  $i \in H_F$ , it must fix  $i$ . Conversely, a lift  $\widehat{\sigma}$  preserves  $H_F$  by order preservation. Thus its stabilizer consists exactly of the lifts  $\widehat{\sigma}$ . There is no compatible ordered-field structure on  $F(i)$  itself, since  $i^2 = -1$ .

## 3 Normal forms, valuation, and summability

Conway normal form identifies the ordered field **No** with real Hahn series having surreal exponents and set-sized well-ordered support. Taking real and imaginary parts coefficientwise gives

$$\mathbf{No} = \mathbb{R}((t^\Gamma)), \quad \mathbb{K} = \mathbb{C}((t^\Gamma)), \quad \Gamma = (\mathbf{No}, +, <). \quad (3.1)$$

The displayed equalities mean canonical identifications. A surcomplex normal form is

$$z = \sum_{g \in S} z_g t^g, \quad z_g \in \mathbb{C}, \quad S \subseteq \Gamma \text{ a well-ordered set.}$$

For  $z \neq 0$ , define

$$v(z) = \min \operatorname{supp}(z), \quad \operatorname{lc}(z) = z_{v(z)}, \quad \operatorname{lt}(z) = \operatorname{lc}(z)t^{v(z)},$$

and set  $v(0) = \infty$ . Multiplication is convolution, with only finitely many contributions to any fixed exponent. Its basic laws are

$$v(zw) = v(z) + v(w), \quad v(z+w) \geq \min\{v(z), v(w)\}.$$

Conjugation acts coefficientwise. Since  $\text{lc}(z\bar{z}) = |\text{lc}(z)|_{\mathbb{C}}^2 > 0$ ,

$$v(\bar{z}) = v(z), \quad v(Nz) = 2v(z), \quad v(|z|) = v(z). \quad (3.2)$$

The valuation ring and its maximal ideal are

$$\mathcal{O} = \{z : v(z) \geq 0\}, \quad \mathfrak{m} = \{z : v(z) > 0\}.$$

They are exactly the finite surcomplex numbers and the infinitesimals, respectively. Every  $z \in \mathcal{O}$  has a unique decomposition  $z = a + \varepsilon$  with  $a \in \mathbb{C}$ ,  $\varepsilon \in \mathfrak{m}$ , and  $\text{st}(z) = a = z_0$ . Thus  $\mathcal{O}/\mathfrak{m} \simeq \mathbb{C}$ . The analogous real residue field is  $\mathbb{R}$ .

**Definition 3.1** (Strong summation). A set-indexed family  $(z_j)_{j \in J}$  is Hahn-summable if the union of its supports is well-ordered and every exponent occurs in only finitely many of the supports. Its sum is then defined coefficientwise by finite sums. An additive map is *strongly additive* when it carries summable families to summable families and commutes with their sums. An automorphism is *strong* here when it and its inverse are strongly additive. A strongly  $\mathbb{C}$ -linear map is a strongly additive  $\mathbb{C}$ -linear map; for an automorphism we also require its inverse to be strongly additive. For field automorphisms,  $\mathbb{C}$ -linearity is equivalent to fixing  $\mathbb{C}$  pointwise.

This definition is a support condition, not a limit of finite partial sums. For example,  $\sum_{n \geq 0} t^n$  is a legitimate Hahn sum and equals  $(1 - t)^{-1}$  even in a setting where its partial sums do not converge in the full fine topology. Section 7 explains that distinction.

**Lemma 3.2** (Support control used below). *If  $S$  is a well-ordered subset of an ordered abelian group and  $\delta > 0$ , then  $S + \mathbb{N}\delta$  is well-ordered, and each exponent has only finitely many representations  $s + n\delta$  with  $s \in S$  and  $n \in \mathbb{N}$ . More generally, the sum of two well-ordered subsets is well-ordered and has finitely many decompositions of any fixed element.*

*Proof.* Every infinite sequence in a well-order has an infinite nondecreasing subsequence. A strictly decreasing sequence of sums, by successively taking such subsequences for its two coordinates, would have a nondecreasing subsequence, a contradiction. Thus the sum is well-ordered. Infinitely many distinct representations of one value would give infinitely many distinct first coordinates. Choosing a strictly increasing subsequence of these forces a strictly decreasing sequence of second coordinates, again impossible. Apply this to  $S$  and  $\mathbb{N}\delta$ .  $\square$

These finite-decomposition arguments justify the rearrangements in the constructions below. More general support-controlled formal exponentials and derivations are developed in [6]; we will need only explicit fixed-shift cases, which can be checked directly.

## 4 Explicit monomial automorphisms and phase twists

Let  $\rho \in \text{Aut}(\mathbb{C})$ , let  $\tau \in \text{Aut}_{\text{ord}}(\Gamma)$ , and let  $\chi : (\Gamma, +) \rightarrow (\mathbb{C}^\times, \cdot)$  be a group homomorphism. Define

$$M_{\rho, \tau, \chi} \left( \sum_g z_g t^g \right) = \sum_g \rho(z_g) \chi(g) t^{\tau(g)}. \quad (4.1)$$

Here  $\chi$  need not arise from an exponential of an additive complex-valued map. Arbitrary multiplicative characters are allowed.

**Theorem 4.1** (Monomial construction). *Formula (4.1) defines a strong field automorphism of  $\mathbb{K}$ , with*

$$v(Mz) = \tau(vz), \quad \text{lc}(Mz) = \rho(\text{lc } z)\chi(vz).$$

*For composition in the order  $M_1M_2$ , the parameters are*

$$\rho = \rho_1\rho_2, \quad \tau = \tau_1\tau_2, \quad \chi(g) = \rho_1(\chi_2(g))\chi_1(\tau_2(g)). \quad (4.2)$$

*The map preserves the real axis exactly when*

$$\rho \in \{\text{id}_{\mathbb{C}}, c|_{\mathbb{C}}\}, \quad \chi(g) \in \mathbb{R}_{>0} \quad \text{for every } g. \quad (4.3)$$

*Proof.* The order bijection  $\tau$  takes a well-ordered support to a well-ordered support. All transformed coefficients on the support are nonzero. Additivity is coefficientwise. For multiplication, each coefficient is a finite sum,  $\tau(g+h) = \tau(g) + \tau(h)$ , and  $\chi(g+h) = \chi(g)\chi(h)$ , so convolution is preserved. The same formula with inverse parameters gives a two-sided inverse; explicitly one can use  $\rho^{-1}, \tau^{-1}$  and  $g \mapsto \rho^{-1}(\chi(\tau^{-1}g)^{-1})$ . Support unions and their finite incidence multiplicities are preserved, proving strongness in both directions. Substitution proves (4.2).

If the real axis is preserved, the image of each real constant is a real constant, so  $\rho(\mathbb{R}) = \mathbb{R}$ . A field automorphism of  $\mathbb{R}$  is the identity, because order and rational cuts determine each real number. Consequently  $\rho$  is identity or conjugation on  $\mathbb{C}$ . Also  $M(t^g)$  is real, so  $\chi(g) \in \mathbb{R}^\times$ . Divisibility of  $\Gamma$  gives  $\chi(g) = \chi(g/2)^2 > 0$ . Conversely, (4.3) makes the image of every real-coefficient series real, and the same holds for the inverse.  $\square$

## 4.1 Rescaling all valuation exponents

For every positive surreal  $a$  let

$$S_a \left( \sum_g z_g t^g \right) = \sum_g z_g t^{ag}. \quad (4.4)$$

Multiplication by  $a$  is an ordered additive automorphism of  $\Gamma$ , so theorem 4.1 applies. We obtain

$$S_a S_b = S_{ab}, \quad S_a^{-1} = S_{1/a}, \quad S_a(t) = t^a, \quad S_a(\omega) = \omega^a.$$

Every  $S_a$  fixes ordinary  $\mathbb{C}$  pointwise, commutes with  $c$ , and preserves order on  $\mathbf{No}$ . It fixes individual valuation values only for  $a = 1$ . For instance,

$$S_2 \left( 3 + 2t^{-1} + (1+i)t^{1/2} \right) = 3 + 2t^{-2} + (1+i)t.$$

This proves the first strict inclusion in (1.1).

Notice that  $S_a$  generally does *not* commute with the omega-map. Indeed,  $S_a(1) = 1$  whereas  $S_a(\omega^1) = \omega^a \neq \omega^{S_a(1)}$  when  $a \neq 1$ . The omega-map must not be mistaken for an exponent reindexing convention.

## 4.2 Positive real characters

For an additive map  $\lambda : \Gamma \rightarrow \mathbb{R}$ , define

$$D_\lambda \left( \sum_g z_g t^g \right) = \sum_g z_g e^{\lambda(g)} t^g.$$

These are strongly  $\mathbb{C}$ -linear real-axis-preserving automorphisms fixing all valuation values. Their group law is  $D_\lambda D_\mu = D_{\lambda+\mu}$ . They generally change leading coefficients, so they are not leading-term-fixing automorphisms.

There are explicit nonzero  $\lambda$ . Regard an exponent  $g$  itself as a surreal normal form and set

$$\ell(g) = [g]_0, \quad (4.5)$$

its real coefficient at exponent zero. This is an additive map defined on all of  $\mathbf{No}$ , and  $\ell(r) = r$  for every real  $r$ . It is not a standard-part map on all surreals: for an infinite  $g$ , coefficient extraction still makes sense even though standard part does not.

### 4.3 Complex phases move the real axis

For  $\theta \in \mathbb{R}$  use the same  $\ell$  to define

$$P_\theta \left( \sum_g z_g t^g \right) = \sum_g z_g \exp(i\theta \ell(g)) t^g. \quad (4.6)$$

Only the ordinary complex exponential is used in the coefficient.

**Theorem 4.2** (A faithful phase action). *The maps  $P_\theta$  form a faithful action of the additive group  $\mathbb{R}$  by strongly  $\mathbb{C}$ -linear, exactly value-preserving automorphisms of  $\mathbb{K}$ . For  $\theta \neq 0$ ,  $P_\theta$  does not preserve  $\mathbf{No}$ . Moreover,*

$$P_\theta P_\phi = P_{\theta+\phi}, \quad c P_\theta c = P_{-\theta}. \quad (4.7)$$

In particular,

$$P_{\pi/2}(t) = it.$$

*Proof.* The character identity follows from additivity of  $\ell$ , so the monomial theorem gives all field and summability assertions. If  $P_\theta = \text{id}$ , then  $e^{i\theta r} = 1$  for every real  $r$ , because  $\ell(r) = r$ . For  $\theta \neq 0$ , taking  $r = \pi/\theta$  contradicts this. Thus the action is faithful. Its parameter is not an angle modulo  $2\pi$ : all real exponents occur, not just integer exponents.

If  $\theta \neq 0$ , the choice  $r = \pi/(2\theta)$  yields  $P_\theta(t^r) = it^r$ , which is not real. Finally, conjugation reverses every coefficient phase, giving (4.7).  $\square$

This provides the second strict inclusion in (1.1) without a choice of transcendence basis or a nonconstructive extension. It also proves that none of the following conditions by itself recovers the real axis: fixing ordinary  $\mathbb{C}$ , strong  $\mathbb{C}$ -linearity, exact valuation preservation, or even all three together.

Valuation preservation is much weaker than exact modulus preservation. For example,

$$|1+t| = 1+t, \quad |P_{\pi/2}(1+t)| = |1+it| = \sqrt{1+t^2} \neq 1+t.$$

Likewise,  $P_{\pi/2}$  does not preserve the distinguished unit circle, by theorem 2.4. Algebraic multiplication survives while the chosen Euclidean geometry changes.

### 4.4 A continuum of explicit conjugations

Set

$$j_\theta = P_\theta c P_\theta^{-1} = P_{2\theta} c, \quad F_\theta = P_\theta(\mathbf{No}). \quad (4.8)$$

Then  $j_\theta^2 = \text{id}$ ,  $\text{Fix}(j_\theta) = F_\theta$ , and  $\mathbb{K} = F_\theta(i)$ . Each  $F_\theta$  is a real closed subfield isomorphic to  $\mathbf{No}$ , but the actual embedded real axes are pairwise distinct. Indeed, equality  $F_\theta = F_\phi$  would make  $P_{\theta-\phi}$  preserve  $\mathbf{No}$ , forcing  $\theta = \phi$ . The fixed-field condition is especially concrete:

$$z \in F_\theta \iff z_g e^{-i\theta \ell(g)} \in \mathbb{R} \text{ for all } g. \quad (4.9)$$

These are different *placements* of an isomorphic real form. We will also construct a real form not isomorphic to  $\mathbf{No}$  in Section 10.

## 5 Leading-term kernels and infinitesimal motion

Define

$$\mathcal{U} = \{\alpha \in \text{Aut}(\mathbb{K}) : v(\alpha z - z) > v(z) \text{ for all } z \neq 0\}. \quad (5.1)$$

Here  $v(0) = \infty$ , so fixed inputs satisfy the inequality. Equivalently,  $\alpha$  fixes the complete leading term of every nonzero series. Such maps are often called 1-automorphisms. They can still change very substantial lower-order data. Let  $\mathcal{U}_{\mathbb{R}}$  denote the corresponding leading-term-fixing subgroup of  $\text{Aut}(\mathbf{No})$ .

### 5.1 A real automorphism need not fix ordinary reals

An order-preserving automorphism  $\sigma$  of  $\mathbf{No}$  fixes every rational. For  $r \in \mathbb{R}$ , rational comparisons imply only

$$\sigma(r) - r \in \mathfrak{m} \cap \mathbf{No},$$

not, for a general  $r$ ,  $\sigma(r) = r$ . For an *irrational* real  $r$ , every infinitesimal perturbation has the same strict rational comparisons: each nonzero ordinary distance  $|r - q|$ ,  $q \in \mathbb{Q}$ , dominates every infinitesimal. Rational  $r$  are fixed already, and real algebraic  $r$  are fixed by their polynomial and order data (lemma 2.1). The following construction makes the distinction explicit.

**Theorem 5.1** (Taylor motion of coefficient constants). *Let  $d : \mathbb{C} \rightarrow \mathbb{C}$  be a field derivation and let  $\delta \in \Gamma$  be positive. Then*

$$\Phi_d \left( \sum_g z_g t^g \right) = \sum_{g, n \geq 0} \frac{d^n(z_g)}{n!} t^{g+n\delta} \quad (5.2)$$

*defines a strong automorphism belonging to  $\mathcal{U}$ , with inverse  $\Phi_{-d}$ . It fixes all monomials  $t^g$ . If  $d(\mathbb{R}) \subseteq \mathbb{R}$  and  $d(i) = 0$ , it preserves  $\mathbf{No}$  and commutes with conjugation. There exist choices for which a real transcendental  $b$  satisfies*

$$\Phi_d(b) = b + t^\delta. \quad (5.3)$$

*Proof.* For an input with support  $S$ , all output exponents lie in  $S + \mathbb{N}\delta$ . Lemma 3.2 proves well-ordering and finite incidence at every exponent. The same argument works for a summable family of inputs and for all double sums needed in a composition. The iterated Leibniz rule

$$d^n(ab) = \sum_{k=0}^n \binom{n}{k} d^k(a) d^{n-k}(b)$$

proves multiplicativity after division by  $n!$ . Additivity is immediate. In  $\Phi_d \Phi_{-d}$ , the coefficient multiplying the  $k$ th derivative at total shift  $k\delta$  is

$$\sum_{n=0}^k \frac{(-1)^n}{n!(k-n)!},$$

which is 1 for  $k = 0$  and 0 otherwise. This proves the inverse formula. Every term with  $n > 0$  has exponent strictly larger than the original minimum; the  $n = 0$  term retains its original leading coefficient. Hence  $\Phi_d \in \mathcal{U}$ . Since  $d(1) = 0$ , each monomial is fixed. For any  $z \neq 0$ , the correction in fact satisfies  $v(\Phi_d(z) - z) \geq v(z) + \delta$ , with  $v(0) = \infty$ . Also  $d(i) = 0$  is automatic from  $d(i^2 + 1) = 2i d(i) = 0$ ; preservation of the real coefficients is the extra condition in the real-axis assertion.

To obtain (5.3), choose a transcendence basis  $B$  of  $\mathbb{R}/\mathbb{Q}$  and an element  $b \in B$ . Define a derivation on  $\mathbb{Q}(B)$  by  $d(b) = 1$  and  $d(b') = 0$  for  $b' \neq b$ . In characteristic zero it extends uniquely through an algebraic extension: for a minimal polynomial  $p(X)$  and a root  $a$ ,

$$d(a) = -\frac{p^d(a)}{p'(a)},$$

where  $p^d$  means differentiate the coefficients. Extend first to  $\mathbb{R}$ , then to  $\mathbb{C}$  by  $d(i) = 0$ . Now  $d^2(b) = 0$ , giving (5.3). All real coefficients stay real, and the construction commutes with  $c$ .  $\square$

Thus even a strongly additive, leading-term-fixing surreal automorphism can move an ordinary real constant. It is not strongly  $\mathbb{R}$ -linear. Confusing strong additivity with coefficient linearity would erase this entire type of automorphism.

The construction uses choice only for a derivation on the *set*  $\mathbb{R}$ ; no class-sized basis is selected. Its action on a given normal form is then an explicit support-controlled formula. It is generally noncomputable from ordinary numerical names of the real coefficients, as should be expected from its use of a transcendence basis.

## 5.2 Strongly complex-linear flows

There are also abundant nontrivial members of  $\mathcal{U}$  fixing all constants. Let  $\lambda : \Gamma \rightarrow \mathbb{C}$  be additive, and fix  $\delta > 0$ . Define

$$\mathcal{D} \left( \sum_g z_g t^g \right) = \sum_g z_g \lambda(g) t^{g+\delta}. \quad (5.4)$$

The positive shift is important:  $g - \delta$  would generally lower valuation, not raise it.

**Theorem 5.2** (A fixed-shift flow). *The map  $\mathcal{D}$  is a strongly  $\mathbb{C}$ -linear derivation, and for every  $s \in \mathbb{C}$ ,*

$$T_s = \sum_{n \geq 0} \frac{s^n \mathcal{D}^n}{n!} \quad (5.5)$$

*is a strongly  $\mathbb{C}$ -linear automorphism in  $\mathcal{U}$ . The group law is  $T_s T_r = T_{s+r}$ . If  $\lambda$  is real-valued and  $s \in \mathbb{R}$ , then  $T_s$  commutes with  $c$  and preserves **No**.*

*Proof.* Additivity of  $\lambda$  proves the Leibniz rule on monomials, and finite convolution proves it on all series. Iterating gives

$$\mathcal{D}^n(t^g) = \left( \prod_{k=0}^{n-1} \lambda(g + k\delta) \right) t^{g+n\delta}.$$

All supports in (5.5) lie in  $S + \mathbb{N}\delta$ . The support lemma justifies summation and the finite regroupings at each coefficient. The iterated Leibniz identity proves multiplicativity, and the binomial formula gives  $T_s T_r = T_{s+r}$ . Thus  $T_{-s}$  is the inverse. Nonconstant terms of the operator exponential strictly increase every input's valuation, so the leading term is unchanged. Constants are killed by  $\mathcal{D}$  because  $\lambda(0) = 0$ . The real-coefficient assertion is immediate.  $\square$

For a particularly transparent example choose  $\delta = 1$  and  $\lambda = \ell$  from (4.5). Then

$$\mathcal{D}(t) = t^2, \quad \mathcal{D}^n(t) = n! t^{n+1}, \quad T_s(t) = \frac{t}{1-st}.$$

Consequently  $T_1$  is a nonidentity strongly  $\mathbb{C}$ -linear, real-axis-preserving 1-automorphism. These explicit flows illustrate the automorphism–derivation connection without requiring a blanket assertion that every formally written operator exponential is summable.

### 5.3 What monomial prescriptions do not automatically give

For a strongly  $\mathbb{C}$ -linear element  $\alpha \in \mathcal{U}$ , the assignments

$$u_g = t^{-g} \alpha(t^g) \in 1 + \mathfrak{m}$$

form a group homomorphism  $\Gamma \rightarrow 1 + \mathfrak{m}$ . Formally one would like to write

$$\alpha \left( \sum_g z_g t^g \right) = \sum_g z_g t^g u_g.$$

This is valid for an existing strong automorphism. Conversely, an arbitrary homomorphism  $g \mapsto u_g$  does not by itself certify that this formula is summable for *every* well-ordered support or that it is onto. One must check the transformed support family and construct a compatible inverse. The fixed-shift proofs above supply exactly those missing checks. This is a genuine obstacle in turning a group decomposition into a complete explicit parametrization of its largest kernel.

## 6 Decomposing the valued automorphism group

Define

$$G_v = \{\alpha \in \text{Aut}(\mathbb{K}) : \alpha(\mathcal{O}) = \mathcal{O}\}.$$

Preserving the valuation here means preserving its comparison structure, not necessarily each value. Such an  $\alpha$  induces

$$\rho_\alpha \in \text{Aut}(\mathbb{C}), \quad \tau_\alpha \in \text{Aut}_{\text{ord}}(\Gamma), \quad v(\alpha z) = \tau_\alpha(vz).$$

The residue action is  $\rho_\alpha(\text{st } z) = \text{st}(\alpha z)$  for finite  $z$ . This does not assert  $\alpha(\mathbb{C}) = \mathbb{C}$ : the coefficient-motion examples have identity residue action while sending constants out of the constant field.

**Theorem 6.1** (Four layers for surcomplex valued automorphisms). *With  $\mathcal{U}$  as in (5.1), every  $\alpha \in G_v$  factors uniquely as*

$$\alpha = u D_\chi M_{\rho, \tau}, \tag{6.1}$$

where  $u \in \mathcal{U}$ ,  $\chi \in \text{Hom}(\Gamma, \mathbb{C}^\times)$ , and

$$D_\chi \left( \sum z_g t^g \right) = \sum z_g \chi(g) t^g, \quad M_{\rho, \tau} \left( \sum z_g t^g \right) = \sum \rho(z_g) t^{\tau(g)}.$$

(In the notation of (4.1),  $D_\chi = M_{\text{id}, \text{id}, \chi}$  and  $M_{\rho, \tau} = M_{\rho, \tau, 1}$ ; the maps  $D_\lambda$  of Section 4 are the  $D_\chi$  with  $\chi = \exp \circ \lambda$ .) In group notation,

$$G_v \simeq (\mathcal{U} \rtimes \text{Hom}(\Gamma, \mathbb{C}^\times)) \rtimes (\text{Aut}(\mathbb{C}) \times \text{Aut}_{\text{ord}}(\Gamma)). \tag{6.2}$$

The outer action on the character factor is

$$((\rho, \tau) \cdot \chi)(g) = \rho(\chi(\tau^{-1}g)). \tag{6.3}$$

The other actions are conjugation by the displayed lifts.

*Proof.* Stabilizing  $\mathcal{O}$  also stabilizes its units and its maximal ideal, so it induces the residue automorphism  $\rho_\alpha$ . Define directly  $\tau_\alpha(g) = v(\alpha(t^g))$ . If  $v(z) = g$ , then  $z/t^g$  is a unit, so  $v(\alpha z) = \tau_\alpha(g)$ . This proves independence from the chosen representative. Multiplication proves additivity; membership of quotients in  $\mathcal{O}$  proves order preservation; the same construction for  $\alpha^{-1}$  supplies the inverse. Thus no quotient whose elements would be

proper classes needs to be formed. The maps  $M_{\rho,\tau}$  give a section of the induced-action homomorphism, by the monomial construction. Let  $I$  be its kernel and put  $\beta = \alpha M_{\rho,\tau}^{-1} \in I$ .

For  $\beta \in I$ , define

$$\chi_\beta(g) = \text{lc}(\beta(t^g)).$$

Since  $\beta$  fixes values, multiplicativity of leading coefficients gives  $\chi_\beta(g+h) = \chi_\beta(g)\chi_\beta(h)$ . For an arbitrary nonzero  $z$ , write  $z = at^g(1+\varepsilon)$  with  $a = \text{lc } z$  and  $v\varepsilon > 0$ . Identity residue action gives  $\text{lc}(\beta a) = a$ , and  $\beta(1+\varepsilon)$  still has leading coefficient 1. Hence

$$\text{lc}(\beta z) = \text{lc}(z)\chi_\beta(vz). \quad (6.4)$$

It follows that  $\beta \mapsto \chi_\beta$  is a group homomorphism on  $I$ , with kernel exactly  $\mathcal{U}$ . The maps  $D_\chi$  split it. Thus  $u = \beta D_{\chi_\beta}^{-1}$  gives (6.1), and all parameters are forced by the preceding operations. More explicitly, the character in that factorization is  $\chi(g) = \text{lc}(\alpha(t^{\tau^{-1}(g)}))$ ; it is indexed after undoing the value action. The subgroup  $\mathcal{U}$  is normal in  $G_v$ , since conjugation by a valued automorphism transports its strict valuation inequality through the ordered map  $\tau$ . Evaluating  $M_{\rho,\tau} D_\chi M_{\rho,\tau}^{-1}$  on a monomial gives (6.3).  $\square$

This direct argument is the full-Hahn instance of the valued-field structure studied in [3]. It also clarifies its limits: (6.2) is not a decomposition of *all* surcomplex field automorphisms. We will exhibit automorphisms outside  $G_v$  in Section 9.

The same proof works with the strong subgroup, replacing  $\mathcal{U}$  by its strong part. All explicit sections are strong, so strength of an automorphism is concentrated in the residual kernel factor. The factor  $\mathcal{U}$  itself is not replaced by a conveniently finite list of coefficients. A strong field map is determined by its action on *both* ordinary coefficients and monomials, using  $\alpha(\sum z_g t^g) = \sum \alpha(z_g) \alpha(t^g)$ . Monomial images alone suffice when it also fixes  $\mathbb{C}$  pointwise. Strongness alone does not remove the coefficient data: the nonidentity Taylor motions [theorem 5.1](#) fix every monomial. Without strongness, the displayed reconstruction is not licensed: Section 7 gives an automorphism that fixes every summand of a Hahn family but moves its sum.

## 6.1 The real reduction

Every field automorphism of **No** preserves order, hence preserves the natural valuation ring. Its residue action on  $\mathbb{R}$  is an ordered-field automorphism and therefore identity. Applying the same proof over  $\mathbb{R}$  gives

$$\text{Aut}(\mathbf{No}) \simeq (\mathcal{U}_{\mathbb{R}} \rtimes \text{Hom}(\Gamma, \mathbb{R}_{>0})) \rtimes \text{Aut}_{\text{ord}}(\Gamma). \quad (6.5)$$

Every character into  $\mathbb{R}^\times$  is positive because  $\Gamma$  is divisible. Applying the ordinary logarithm pointwise identifies its group with  $\text{Hom}(\Gamma, \mathbb{R})$  under addition.

Combining (6.5) with the real-axis theorem gives a structural description of  $G_{\mathbf{No}}$ :

$$G_{\mathbf{No}} \simeq [(\mathcal{U}_{\mathbb{R}} \rtimes \text{Hom}(\Gamma, \mathbb{R}_{>0})) \rtimes \text{Aut}_{\text{ord}}(\Gamma)] \times C_2. \quad (6.6)$$

In particular, “residue action is identity” must not be shortened to “all embedded real constants are fixed.” [Theorem 5.1](#) is an explicit counterexample to that shortcut.

## 6.2 One level deeper: the ordered additive value group

Because  $\Gamma = (\mathbf{No}, +, <)$  is itself the ordered additive group of real Hahn series, one can further analyze  $\text{Aut}_{\text{ord}}(\Gamma)$ . Let  $\mathcal{U}_{\text{add}}$  consist of its additive automorphisms fixing each leading term. For an order bijection  $\zeta : \Gamma \rightarrow \Gamma$  and positive real scalars  $(a_g)_{g \in \Gamma}$ , the formula

$$A_{\zeta,a} \left( \sum_g r_g t^g \right) = \sum_g a_g r_g t^{\zeta(g)} \quad (6.7)$$

defines an ordered additive automorphism. Here  $\zeta$  need not be additive: this is an additive-group construction, not a field construction.

An ordered additive automorphism  $\tau$  permutes the Archimedean classes in order. Its parameters can be read directly as  $\zeta(g) = v(\tau(t^g))$  and  $a_g = \text{lc}(\tau(t^g)) > 0$ . For fixed  $g$ , comparison of  $rt^g$  with rational multiples of  $t^g$  shows that the leading coefficient of  $\tau(rt^g)$  is  $a_g r$  for every  $r \in \mathbb{R}^\times$ , at value  $\zeta(g)$ . A tail of value greater than  $g$  is infinitesimal relative to  $t^g$  in the ordered additive group, and its image has value greater than  $\zeta(g)$ . Thus the leading term of the image of any input with leading term  $rt^g$  is  $a_g r t^{\zeta(g)}$ . This uses only order and additivity, not strong summation or a quotient whose elements would be proper classes. Removing (6.7) leaves exactly  $\mathcal{U}_{\text{add}}$ , and the parameters are unique. This is the additional kernel–skeleton factorization, parallel to [4].

It is important to retain the reindexing action in its group law. Direct calculation gives

$$A_{\zeta,a} A_{\eta,b} = A_{\zeta\eta,d}, \quad d_g = a_{\eta(g)} b_g. \quad (6.8)$$

Thus these parameters have a semidirect-product law, not componentwise multiplication of an unrelated permutation and weight family. The ordinary logarithm turns positive weights into additive real weights but does not remove their reindexing. Together with (6.5), this identifies five structural layers on the real side: field leading-term motion, characters, additive leading-term motion, exponent-order permutations, and Archimedean coefficient weights.

## 7 Fine topology does not mean strong summation

The fine topology on  $\mathbb{K}$  is generated by the balls

$$B(a, r) = \{z : |z - a| < r\}, \quad r \in \mathbf{No}_{>0}.$$

By (3.2), it is the same topology as the valuation topology with neighborhoods

$$V_\gamma(a) = \{z : v(z - a) > \gamma\}, \quad \gamma \in \Gamma.$$

For example,  $V_{v(r)}(a) \subseteq B(a, r)$ , while a ball of radius  $t^{\gamma+1}$  is contained in  $V_\gamma(a)$ . These inclusions suffice to identify the neighborhood systems.

**Proposition 7.1.** *Every element of  $G_v$  is a fine homeomorphism and a uniform equivalence. In particular, every real-axis-preserving surcomplex field automorphism is fine-continuous.*

*Proof.* If  $\alpha$  induces  $\tau$  on  $\Gamma$ , then

$$v(\alpha z - \alpha a) = \tau(v(z - a)).$$

The order bijection  $\tau$  takes the neighborhood basis to itself: for a target threshold  $\eta$ , use the input threshold  $\tau^{-1}(\eta)$ . This threshold is independent of the base point. Applying the same equation to pairs of points proves uniform continuity of both  $\alpha$  and its inverse. The real-axis subgroup lies in  $G_v$  by lemma 2.1 and (2.4).  $\square$

This is an inclusion in the continuous automorphism group, not a classification of that whole group. In particular, we do not infer a converse merely from continuity. The chosen valuation ring contains additional information beyond the neighborhood calculation used here.

The phase twists are therefore continuous even though they do not preserve the real axis. Lifting any ordinary field automorphism  $\rho$  of  $\mathbb{C}$  coefficientwise gives further examples in  $G_v$ . There is no contradiction with the familiar classification of continuous automorphisms of ordinary  $\mathbb{C}$ : the topology induced on  $\mathbb{C} \subset \mathbb{K}$  is *discrete*, not its usual Archimedean topology. A ball of infinitesimal radius around a constant contains no other ordinary complex constant.

## 7.1 Why small nets cannot test the full topology

**Proposition 7.2.** *Every set-sized subset of the full surcomplex field is closed and discrete. Every convergent set-indexed net is eventually equal to its limit. Nevertheless, the full field has no isolated points.*

*Proof.* For a set  $S \subseteq \mathbb{K}$  and a point  $a$ , the values  $\{v(s - a) : s \in S, s \neq a\}$  form a set of surreal numbers. Every set of surreals has an upper bound, so choose  $\gamma$  strictly above all of them. Then  $V_\gamma(a)$  misses  $S \setminus \{a\}$ . This proves both closedness and discreteness. Apply the same argument to the range of a set-indexed net together with its limit to get eventual equality. Finally,  $a + t^{\gamma+1} \neq a$  belongs to  $V_\gamma(a)$ , so no point of the full field is isolated.  $\square$

For example,  $t^n$  does not tend to zero in the full surreal fine topology: none of these terms lies in  $V_\omega(0)$ . In the fixed field  $\mathbb{C}((t^\mathbb{Q}))$ , by contrast, the integers are cofinal in its value group and  $t^n \rightarrow 0$  in its intrinsic valuation topology. A set-sized Hahn workspace inside  $\mathbb{K}$  acquires a discrete subspace topology from the full field, even when its intrinsic valuation topology is nondiscrete. The choice of topology is part of the theorem, not merely a presentational convention.

## 7.2 A continuous automorphism that fails to preserve a Hahn sum

We give an explicit support-correct witness. We will use the standard Newton–Puiseux fact that

$$\mathcal{P} = \bigcup_{m \geq 1} \mathbb{R}((t^{1/m}))$$

is real closed. Indeed,  $\mathcal{P}(i) = \bigcup_{m \geq 1} \mathbb{C}((t^{1/m}))$  is algebraically closed by Newton–Puiseux [14]; an ordered field with algebraically closed quadratic extension by  $i$  is real closed. This set-sized Puiseux field is also discussed among the special cases in [3, Section 5.5]. Define

$$a = \sum_{n \geq 2} t^{1-1/n}, \quad \varepsilon = t^\omega. \quad (7.1)$$

The support  $\{1/2, 2/3, 3/4, \dots\}$  is a well-ordered increasing sequence. Its denominators are unbounded, so  $a \notin \mathcal{P}$ . Since  $\mathcal{P}$  is real closed,  $a$  is transcendental over  $\mathcal{P}$ .

For each  $p \in \mathcal{P}$ , the nonzero difference  $a - p$  has rational support and hence rational valuation. Thus  $\varepsilon$  is smaller than  $|a - p|$ . It follows that  $a$  and  $a + \varepsilon$  have the same cut over  $\mathcal{P}$ . An ordered isomorphism  $\mathcal{P}(a) \rightarrow \mathcal{P}(a + \varepsilon)$  fixing  $\mathcal{P}$  and moving  $a$  therefore exists; real closedness of  $\mathcal{P}$  ensures that signs of rational functions are determined by this cut. Extend it through real closures and then to an automorphism  $\sigma$  of  $\mathbf{No}$  by the ordered extension theorem proved in Section 9.

Every summand in (7.1) belongs to  $\mathcal{P}$  and is fixed. But

$$\sigma(a) = a + \varepsilon \neq a = \sum_{n \geq 2} \sigma(t^{1-1/n}).$$

The lift of  $\sigma$  to  $\mathbb{K}$  fixes ordinary  $\mathbb{C}$ , commutes with conjugation, and is fine-continuous, but is not strong. We have not claimed that this particular extension fixes every value or every leading term; those stronger assertions are unnecessary here.

There is also a finite-observation obstruction to approximation by strong maps. Any real surreal automorphism fixing  $t$  fixes all its positive rational powers, by uniqueness of positive roots. If it is strong, it fixes  $a$ . Hence the neighborhood specified by

$$\psi(t) = t, \quad \psi(a) = a + \varepsilon$$

in the pointwise-equality topology on  $\text{Aut}(\mathbf{No})$  contains  $\sigma$  and contains no strong automorphism. This proves non-density in that particular automorphism-group topology, not in every possible topology on the group.

The same finite observations exclude strong maps in the full  $\text{Aut}(\mathbb{K})$ . If  $\psi(t) = t$  and  $q_n = 1 - 1/n$ , then  $\psi(t^{q_n})^n = t^{n-1}$ , so  $\psi(t^{q_n}) = \zeta_n t^{q_n}$  with  $\zeta_n$  an ordinary complex  $n$ th root of unity. Strongness would force  $\psi(a) = \sum_{n \geq 2} \zeta_n t^{q_n}$ , whose support is contained in  $\mathbb{Q}$  and cannot contain the exponent  $\omega$  of  $a + \varepsilon$ . The lifted witness satisfies both observations, so the strong subgroup is not dense there either. This argument requires neither preservation of the real axis nor fixation of all ordinary complex constants by  $\psi$ .

## 8 Differentiability: a surprising zero-derivative phenomenon

For a function  $f : \mathbb{K} \rightarrow \mathbb{K}$ , its fine derivative at  $z$  is the punctured-neighborhood limit

$$f'(z) = \lim_{h \rightarrow 0, h \neq 0} \frac{f(z+h) - f(z)}{h},$$

when this limit exists in the full fine topology. The limit quantifies over all small surcomplex increments; it is not tested only on an ordinary sequence. This is the same difference-quotient convention encoded in the repository's generic `FineHasDerivAt` interface [13].

**Theorem 8.1** (Derivative dichotomy for field automorphisms). *If a field automorphism  $\alpha$  of  $\mathbb{K}$  has a fine derivative at one point, it has the same derivative at every point. If that derivative is nonzero, then  $\alpha = \text{id}$  and the derivative is 1. Nonidentity differentiable automorphisms are nevertheless possible: they must have derivative 0 everywhere.*

*Proof.* Additivity makes every difference quotient equal to  $q(h) = \alpha(h)/h$ , independently of the base point. For a fixed  $b \neq 0$ ,

$$q(bh) = \frac{\alpha(b)}{b} q(h).$$

If  $q(h) \rightarrow L$ , multiplication by  $b$  is a homeomorphism of the punctured neighborhood of zero, so the left-hand side also tends to  $L$ . Hausdorff uniqueness gives  $L = \alpha(b)L/b$ . When  $L \neq 0$ , cancellation forces  $\alpha(b) = b$  for every nonzero  $b$ . Thus  $\alpha = \text{id}$ , whose quotient is constantly 1.  $\square$

**Theorem 8.2** (Exponent dilations have derivative zero). *For the maps  $S_a$  of (4.4),*

$$S'_a \equiv 0 \quad (a > 1), \quad S'_1 \equiv 1.$$

*For  $0 < a < 1$ ,  $S_a$  has no  $\mathbb{K}$ -valued fine derivative at any point. In particular,  $S_2$  is a nonidentity field automorphism and a fine homeomorphism with derivative zero everywhere.*

*Proof.* For every nonzero  $h$ ,

$$v\left(\frac{S_a(h)}{h}\right) = (a-1)v(h). \tag{8.1}$$

If  $a > 1$ , given any  $\eta \in \Gamma$ , taking  $v(h) > \eta/(a-1)$  forces the quotient into  $V_\eta(0)$ . Thus the quotient tends to zero. This argument works for positive infinitesimal  $a-1$  as well as for ordinary positive  $a-1$ . If  $0 < a < 1$ , theorem 8.1 would force any derivative to be zero, since  $S_a(t) = t^a \neq t$ . But for every proposed input threshold  $\gamma_0$ , choose  $\gamma > \max\{\gamma_0, 0\}$  and put  $h = t^\gamma$ . Then  $h \in V_{\gamma_0}(0)$ , while the quotient has value  $(a-1)\gamma < 0$  and lies outside  $V_0(0)$ . This rules out a zero limit using the full neighbourhood criterion, without a set-indexed cofinal net.  $\square$

Conjugating the input does not change its valuation. Thus  $S_a c$  also has zero derivative for  $a > 1$ , even though  $c$  by itself is not surcomplex-differentiable. For  $c$ , real and imaginary increments give quotients 1 and  $-1$  at arbitrarily small scales.

**Corollary 8.3.** *If a field automorphism and its inverse are both fine-differentiable, the automorphism is identity.*

*Proof.* Differentiability implies continuity by writing an increment of the function as its difference quotient times the input increment. If both maps were nonidentity and differentiable, both derivatives would be zero. Put  $k = \alpha(h)$ ; continuity implies  $k \rightarrow 0$  as  $h \rightarrow 0$ . The product

$$\frac{\alpha(h)}{h} \frac{\alpha^{-1}(k)}{k} = 1$$

would then tend to zero, a contradiction. Injectivity ensures  $k \neq 0$  for  $h \neq 0$ , so the inverse's punctured limit applies. The identity satisfies both conditions.  $\square$

These results do not assert that  $S_2$  is representable by a locally convergent power series with fixed coefficients. They concern the fine difference-quotient definition. In this setting, “everywhere differentiable” cannot be used as a synonym for ordinary complex analytic rigidity, and a zero derivative does not imply local constancy:  $S_2$  is injective and the full field has no isolated points. The Berarducci–Mantova scalar derivation discussed later is a different operation again.

## 9 The unrestricted field: homogeneity and fixed fields

The preceding explicit formulas already disprove real-axis rigidity. The full pure-field picture is even less rigid. To discuss it without misusing a set-sized extension theorem on a proper class, we give the extension argument directly.

**Theorem 9.1** (Class back-and-forth). *Assume global choice. Let  $K_1, K_2$  be proper-class algebraically closed fields of the same characteristic. Every isomorphism between set-sized subfields of  $K_1$  and  $K_2$  extends to an isomorphism  $K_1 \rightarrow K_2$ . In particular, every isomorphism between set-sized subfields of  $\mathbb{K}$  extends to a field automorphism of  $\mathbb{K}$ .*

*Proof.* Fix a set-like global well-order and the resulting class enumerations of  $K_1$  and  $K_2$ , indexed by **On**. Recursively maintain an isomorphism  $f_\alpha : A_\alpha \rightarrow B_\alpha$  between set-sized subfields. At a forth step, take the next element  $x$  of  $K_1$ . If  $x$  is algebraic over  $A_\alpha$ , send it to a root in  $K_2$  of its transported irreducible polynomial, extending the field isomorphism. If  $x$  is transcendental, choose a transcendental element over  $B_\alpha$ . This is possible because the elements of  $K_2$  algebraic over a set-sized subfield form a set: there are set many polynomials and each has finitely many roots. A proper-class field cannot be exhausted by it.

At each successor stage perform both a forth step and a back step, the latter for the next element of  $K_2$  using the inverse partial map; skip an element already in the relevant partial field. Always select the least admissible witness in the fixed global well-order. At a limit ordinal take unions. There are only set many earlier stages at any given ordinal, so the partial domains and ranges remain sets. The class union of all partial maps is a field isomorphism; forth steps exhaust  $K_1$  and back steps exhaust  $K_2$ . Each partial map is a set. The deterministic rule has a unique recursion of every set ordinal length, with the ambient fields and well-order as class parameters; these compatible recursions define the class union. The candidate witnesses can form a proper class, but its least member is selected by the global well-order. No recursion whose stage value is itself a proper class, or class-sized transcendence basis, is needed.  $\square$

**Theorem 9.2** (Ordered surreal extension). *Under the same foundational convention, every ordered-field isomorphism between set-sized subfields of  $\mathbf{No}$  extends to an automorphism of  $\mathbf{No}$ .*

*Proof.* First extend the map to real closures inside  $\mathbf{No}$ ; uniqueness of ordered real closures supplies the extension. Run a back-and-forth as above, now keeping the partial fields real closed. An element  $x$  outside a partial real closed field  $A$  is transcendental over it. Its comparisons with  $A$  form a set-sized cut  $(\mathcal{L}, \mathcal{R})$  with  $\mathcal{L} \cup \mathcal{R} = A$ . Transport this cut to the current range  $B$ . The surreal cut property supplies an element  $y$  strictly between every element of its left and right sides. It cannot belong to  $B$ , since every element of  $B$  occurs on one of those sides. Thus  $y$  is transcendental over  $B$ .

The substitution  $x \mapsto y$  gives an ordered isomorphism of rational function fields. To see preservation of signs, factor polynomials over the real closed coefficient field into linear factors and quadratic factors without real roots; their signs are determined by the cut and leading coefficient. Extend again to real closures. At limits, unions of increasing real closed fields are real closed, since each polynomial has its finitely many coefficients in one stage. Forth and back exhaust  $\mathbf{No}$ .  $\square$

These proofs explain why field homogeneity is so strong and why it should not be confused with simplicity preservation. The selected separator  $y$  need not have the same birthday or sign-sequence complexity as  $x$ .

## 9.1 Orbits over ordinary complex constants

Every element of  $\mathbb{K} \setminus \mathbb{C}$  is transcendental over  $\mathbb{C}$ , since  $\mathbb{C}$  is algebraically closed. Therefore, for  $x, y \notin \mathbb{C}$ , the substitution  $x \mapsto y$  is an isomorphism  $\mathbb{C}(x) \rightarrow \mathbb{C}(y)$  and extends by theorem 9.1. We obtain

$$\text{Aut}(\mathbb{K}/\mathbb{C}) \text{ is transitive on } \mathbb{K} \setminus \mathbb{C}. \quad (9.1)$$

The analogous statement for finite algebraically independent tuples follows by the same rational-function argument. Thus a pure-field automorphism fixing every ordinary complex number can send an infinite surreal to an infinitesimal, or to an element off the real axis.

**Proposition 9.3.** *For every set  $A \subseteq \mathbb{K}$ , there are automorphisms fixing  $A \cup \mathbb{C}$  pointwise which do not preserve  $\mathbf{No}$ . There are also such automorphisms which do not preserve  $\mathcal{O}$ . Consequently, neither the distinguished real axis nor the natural valuation ring is invariant under all pure-field automorphisms over any prescribed set of parameters.*

*Proof.* Let  $E$  be the algebraic closure inside  $\mathbb{K}$  of the field generated by  $A \cup \mathbb{C}$ . It is a set. Choose a positive infinite  $x \in \mathbf{No} \setminus E$ ; for instance choose  $x$  larger than the moduli of all elements of  $E$ . Then  $x$  is transcendental over  $E$ . Both maps

$$E(x) \longrightarrow E(ix), \quad x \mapsto ix, \quad E(x) \longrightarrow E(x^{-1}), \quad x \mapsto x^{-1}$$

are field isomorphisms fixing  $E$ . Extend each to  $\mathbb{K}$ . The first moves a real element off the real axis. The second sends an element outside  $\mathcal{O}$  into its maximal ideal, so cannot stabilize  $\mathcal{O}$ .  $\square$

In particular, these structures are not first-order definable in the pure field with parameters: definable sets are invariant under automorphisms fixing their parameters. This conclusion is stronger than merely saying that no preferred coordinate system has yet been chosen. Combining the proposition with the phase example gives the proper chain

$$G_{\mathbf{No}} \subsetneq G_v \subsetneq \text{Aut}(\mathbb{K}). \quad (9.2)$$

The first strictness has an explicit strong,  $\mathbb{C}$ -linear, value-fixing witness; the second has a class back-and-forth witness.

## 9.2 Fixed fields of the main collections

Let  $\mathbb{Q}^{\text{rc}} \subset \mathbb{R}$  be the field of real algebraic numbers and let  $\overline{\mathbb{Q}} = \mathbb{Q}^{\text{rc}}(i) \subset \mathbb{C}$ .

**Proposition 9.4.** *In the class-map interpretation,*

$$\begin{aligned} \text{Fix}(\text{Aut}(\mathbb{K})) &= \mathbb{Q}, & \text{Fix}(\text{Aut}(\mathbb{K}/\mathbb{C})) &= \mathbb{C}, \\ \text{Fix}(\text{Aut}(\mathbf{No})) &= \mathbb{Q}^{\text{rc}}, & \text{Fix}(G_{\mathbf{No}}) &= \mathbb{Q}^{\text{rc}}. \end{aligned}$$

*The fixed field of the  $i$ -fixing lifts of all surreal automorphisms is  $\overline{\mathbb{Q}}$ .*

*Proof.* A nonrational algebraic element has a distinct conjugate in characteristic zero, and its conjugacy isomorphism extends by [theorem 9.1](#). A transcendental element can similarly be moved by a rational-function isomorphism. This gives the first assertion. Transitivity (9.1) gives the second.

Every surreal automorphism fixes  $\mathbb{Q}^{\text{rc}}$  by order. For  $x \notin \mathbb{Q}^{\text{rc}}$ ,  $x$  is transcendental over this real closed field. Choose a positive  $\varepsilon$  smaller than all nonzero distances  $|x - a|$  for  $a \in \mathbb{Q}^{\text{rc}}$ . This is possible for a set of distances by the surreal cut property. Then  $x$  and  $x + \varepsilon$  have the same cut, so the ordered extension theorem moves  $x$ . This proves the third assertion. For the last two assertions apply the coordinate formula (2.3); allowing  $c$  kills the imaginary fixed coordinate, whereas forbidding  $c$  leaves both coordinates in  $\mathbb{Q}^{\text{rc}}$ .  $\square$

## 9.3 Why a transcendence-basis slogan is not a classification

For a set-sized algebraically closed field, specifying a transcendence basis is useful for producing automorphisms: first choose compatible images of transcendental generators and then extend through algebraic elements. But an automorphism need not permute one fixed basis, and its action on algebraic elements is additional data. Even there, “the automorphism group is the symmetric group on a transcendence basis” is false.

For the full surcomplex field, the corresponding basis would be a proper class and requires additional foundational care. The set-stage back-and-forth proof is sufficient for all existential statements here. It gives neither a computational procedure on arbitrary surcomplex inputs nor a canonical parametrization of all class automorphisms. The much more explicit decomposition in Section 6 succeeds precisely because it retains a valuation and a normal-form presentation.

## 10 Involutions and genuinely different real forms

An *involution* in this section means an automorphism of order exactly two. A *real form* of  $\mathbb{K}$  means a real closed subfield  $F \subset \mathbb{K}$  such that  $\mathbb{K} = F(i)$ . Its conjugation is the involution fixing  $F$  and sending  $i$  to  $-i$ . The fields  $F_\theta$  from (4.8) already give many examples, all isomorphic to the distinguished  $\mathbf{No}$ .

We use the classical Artin–Schreier theorem: if an algebraically closed field is a proper finite extension of a subfield, that subfield is real closed, the characteristic is zero, and the extension has degree two. A proof and the finite-automorphism application are given in [8]. We apply these ordinary set-field theorems inside invariant set-sized algebraically closed subfields, as follows.

**Theorem 10.1** (Finite subgroups and real forms). *Every nontrivial finite subgroup of  $\text{Aut}(\mathbb{K})$  has order two. Every nonidentity finite-order automorphism is an involution  $j$ , its fixed field  $F_j$  is real closed, and  $\mathbb{K} = F_j(i)$ . In particular,  $j(i) = -i$ . Consequently  $\text{Aut}(\mathbb{K}/\mathbb{C})$  and the leading-term kernel  $\mathcal{U}$  are torsion-free. Within  $G_{\mathbf{No}}$ , the only nonidentity torsion element is  $c$ .*

*Proof.* For a nontrivial finite indexed group  $H$ , choose for each nonidentity element a moved point. Adjoin  $i$  and close this finite set under  $H$ , then take the relative algebraic closure in  $\mathbb{K}$  of its generated field. The result is a set-sized,  $H$ -stable algebraically closed field  $E$ , and the restricted action is faithful. Artin's fixed-field theorem and Artin–Schreier give  $[E : E^H] = |H| = 2$ . Applying this to a finite cyclic action shows that every nonidentity finite-order automorphism is an involution.

For such an involution  $j$ , the same construction can include any prescribed set of elements of  $\text{Fix}(j)$ . Each  $E^j$  is real closed, and these fields form a directed family with union  $F_j = \text{Fix}(j)$ ; hence  $F_j$  is real closed. Because  $i \in E$  and  $E^j$  is real closed,  $j(i) = -i$ . For every  $z \in \mathbb{K}$ ,

$$z = \frac{z + jz}{2} + i \frac{z - jz}{2i},$$

and both displayed coordinates are fixed by  $j$ . Thus  $\mathbb{K} = F_j(i)$ . An automorphism fixing  $\mathbb{C}$  cannot be such an involution. A member of  $\mathcal{U}$  must fix  $i$ , since changing it to  $-i$  changes its leading coefficient, and is therefore also torsion-free. Finally, an order-preserving bijection of a linear order has no nontrivial finite order: a moved point has a strictly monotone finite orbit segment. Use  $G_{\mathbf{No}} \simeq \text{Aut}(\mathbf{No}) \times C_2$ .  $\square$

Two involutions are conjugate in  $\text{Aut}(\mathbb{K})$  exactly when their fixed real fields are isomorphic. One direction follows by restricting a conjugating automorphism. Conversely an isomorphism  $F_j \rightarrow F_k$  extends by  $x + iy \mapsto f(x) + if(y)$ , and this extension conjugates  $j$  to  $k$ . Thus classifying involutions is a real-form classification problem, not merely the observation that  $i$  has two possible images.

## 10.1 A real form not isomorphic to the surreal field

The following construction shows that not all involutions of  $\mathbb{K}$  are conjugate to its distinguished conjugation.

**Theorem 10.2.** *Assuming the class-theoretic framework above,  $\mathbb{K}$  has a real form with a countable cofinal subset. It is not isomorphic as a field to  $\mathbf{No}$ . The associated involution is not conjugate to  $c$ .*

*Proof.* Form the rational function field  $\mathbf{No}(T)$  with  $T$  transcendental, ordered so that  $T$  is larger than every surreal coefficient. The sign of a nonzero rational function is determined by the ratio of its leading coefficients. Let

$$H = \mathbf{No}(T)^{\text{rc}}$$

be an ordered real closure. To construct this in class foundations, choose a set-like enumeration of  $\mathbf{No}(T)$  and increasing set-sized subfields  $A_\alpha$  exhausting it. For each  $\alpha$  choose a set-coded ordered real closure  $R_\alpha$  of  $A_\alpha$ , using the fixed global well-order. For  $\alpha \leq \beta$ , there is a unique ordered embedding  $R_\alpha \rightarrow R_\beta$  extending  $A_\alpha \subseteq A_\beta$ . Uniqueness follows by preserving the order of the finitely many real roots of each polynomial, and makes these embeddings compatible under composition. Take the direct limit: represent an element by the least equivalent stage–element pair in the set-like global well-order. Thus the elements are sets, not proper-class equivalence classes. Every finite tuple occurs in one stage, so the resulting class field is real closed; every element is algebraic over the embedded  $\mathbf{No}(T)$ . Identifying that canonical image with  $\mathbf{No}(T)$  gives the required extension. This is a coherent set-stage construction, not an application of a class Zorn lemma.

The powers  $T, T^2, T^3, \dots$  are cofinal in  $\mathbf{No}(T)$ . Indeed, each rational function has finite degree and each nonzero coefficient and its reciprocal is bounded in absolute value by  $T$ . Increasing an integer power of  $T$  therefore dominates the function. The same powers remain cofinal in  $H$ : if  $h$  is algebraic, a monic polynomial for  $h$  has finitely many coefficients, and the usual algebraic root bound

$$|h| \leq 1 + \max_{k < n} |a_k| \quad \text{when } h^n + a_{n-1}h^{n-1} + \dots + a_0 = 0$$

is valid in any ordered field. Its right-hand side is bounded by a power of  $T$ .

The field  $H$  is proper-class-sized because it contains  $\mathbf{No}$ . Its complexification  $H(i)$  is algebraically closed of characteristic zero. By [theorem 9.1](#), it is isomorphic as a pure field to  $\mathbb{K}$ . Transport  $H$  and its conjugation across this isomorphism. The transported real form has a countable cofinal subset. By contrast, every set-sized subset of  $\mathbf{No}$  is bounded above, so  $\mathbf{No}$  has no countable cofinal subset. A field isomorphism between real closed fields preserves their unique orders; hence these real forms cannot be isomorphic. The conjugacy criterion for involutions proves the final assertion.  $\square$

This separates two phenomena. The phase family gives distinct embedded copies of the *same* real form. [Theorem 10.2](#) gives a different isomorphism type of real form. Neither result supplies a classification of all real closed class fields whose complexification is  $\mathbb{K}$ .

## 11 Simplicity, the omega-map, and exponential structure

The large automorphism groups above forget much of what makes a surreal number canonical. Adding back suitable structure can impose rigidity, but different added structures impose different conditions.

### 11.1 Simplicity restores the canonical surreal numbers

Let  $<_s$  denote proper-prefix simplicity on ordinal-length surreal sign sequences. A bijection preserving both numerical order and simplicity is identity. For completeness, fix the empty sign sequence. At a successor stage, the two immediate simplicity extensions of a fixed sequence lie on opposite numerical sides, so both are fixed. At a limit stage, the unique sign sequence whose proper initial segments are the already fixed chain is fixed. Transfinite induction proves the assertion.

Thus a member of  $G_{\mathbf{No}}$  whose real restriction preserves simplicity is either identity or conjugation; marking  $i$  eliminates conjugation. This is rigidity of an enriched structure, not rigidity of the real closed field reduct. A transformation such as  $S_2$  is allowed to change birthdays and simplicity and therefore does not contradict it.

### 11.2 The omega-map detects exact value preservation

Suppose  $\sigma \in \text{Aut}(\mathbf{No})$  commutes with  $x \mapsto \omega^x$  and fixes valuation values pointwise. Since  $v(\omega^x) = -x$ ,

$$-\sigma(x) = v(\omega^{\sigma(x)}) = v(\sigma(\omega^x)) = v(\omega^x) = -x.$$

Therefore  $\sigma = \text{id}$ . The qualifier “pointwise” is indispensable: all surreal field automorphisms preserve the natural valuation *up to* an ordered group automorphism, and that weaker fact does not give the last equality. Without pointwise fixation, the displayed computation instead says that the induced value-group action equals  $\sigma$  itself on the underlying surreal elements. That is a compatibility condition, not a proof of triviality.

### 11.3 Exponential faithfulness: the existing repository result

The following finite argument is taken from the repository’s *Growth-Scale Rigidity* manuscript [10]. We give the proof because it is short, powerful, and avoids exchanging an automorphism with an infinite sum.

**Lemma 11.1** (Displacement cannot be globally bounded). *Let  $F$  be an ordered field and  $\sigma : F \rightarrow F$  a nonidentity unital field endomorphism. Then  $\{\sigma(x) - x : x \in F\}$  is cofinal and coinital in  $F$ .*

*Proof.* Write  $D(x) = \sigma(x) - x$ . The correct multiplication identity is

$$D(ab) = \sigma(a)D(b) + bD(a). \quad (11.1)$$

Choose  $a$  with  $d = D(a) \neq 0$ . Given  $B > 0$ , put

$$b = \frac{2B(1 + |\sigma(a)|)}{|d|}.$$

If both  $|D(b)|$  and  $|D(ab)|$  were at most  $B$ , then

$$2B(1 + |\sigma(a)|) = |bd| \leq |D(ab)| + |\sigma(a)||D(b)| \leq B(1 + |\sigma(a)|),$$

a contradiction. Thus absolute displacements have no upper bound. Since  $D(-x) = -D(x)$ , both signs occur with arbitrarily large magnitude.  $\square$

**Theorem 11.2** (Faithful value-group action). *Let  $F$  be an ordered field with an increasing group isomorphism  $E : (F, +) \rightarrow (F_{>0}, \cdot)$ , and let  $w : F^\times \rightarrow \Lambda$  be a nontrivial convex valuation. If an automorphism  $\sigma$  commutes with  $E$  and fixes all  $w$ -values, then  $\sigma = \text{id}$ . Hence the exponential automorphisms stabilizing the valuation ring act faithfully on its value group.*

*Proof.* The homomorphism

$$\nu : F \rightarrow \Lambda, \quad \nu(x) = w(E(x))$$

is additive, with image exactly  $w(F^\times)$ : for  $a \neq 0$  use  $x = E^{-1}(|a|)$  and  $w(-1) = 0$  to get  $\nu(x) = w(a)$ . Thus it is onto the actual value group, even if  $\Lambda$  was presented as a larger ambient group. Nontriviality of  $w$  makes its kernel  $H$  a proper convex additive subgroup:  $E$  is increasing and a convex valuation reverses the order on positive elements. A proper convex additive subgroup is bounded above in  $F$ ; if  $b > 0$  lies outside it, then every  $h \in H$  satisfies  $|h| < b$ .

Commutation and pointwise value fixation give, for every  $x$ ,

$$\nu(\sigma x - x) = w(E(\sigma x)) - w(E(x)) = w(\sigma(E(x))) - w(E(x)) = 0.$$

Thus all displacements lie in the bounded subgroup  $H$ . Lemma 11.1 forces  $\sigma = \text{id}$ . Taking the kernel of the induced-action homomorphism proves faithfulness.  $\square$

For the usual surreal exponential, the kernel  $H$  is exactly the finite real-surreal elements. Indeed, bounded  $x$  gives bounded  $\exp x$  and bounded reciprocal by comparison with  $e^n$ ; conversely, if  $\exp x$  and its reciprocal are bounded, comparison with integers and application of  $\log$  bound  $x$ . Consequently,

$$\ker(\text{Aut}_{\exp}(\mathbf{No}) \rightarrow \text{Aut}_{\text{ord}}(\Gamma)) = \{\text{id}\}. \quad (11.2)$$

In particular, a nonidentity surreal 1-automorphism cannot preserve the total real exponential. This settles the precise mathematical alternative posed in Question 5.4 of the inspected arXiv version of [5], by the repository proof above; it is not a new result of this article. Faithfulness does not say that the exponential automorphism group is trivial. It says that no nonidentity member can hide all its motion below the valuation scale.

### 11.4 A simple obstruction to exponential lifting

An ordered automorphism of the value group always has the Hahn lift  $M_{\text{id}, \tau}$ . It need not have *any* lift preserving the total surreal exponential. One elementary instance, also covered in [10], is worth spelling out.

**Proposition 11.3.** *For  $q \in \mathbb{Q}_{>0}$  with  $q \neq 1$ , the value-group dilation  $g \mapsto qg$  is not induced by any exponential automorphism of  $\mathbf{No}$ .*

*Proof.* If  $\sigma$  were such an automorphism, the homomorphism  $v(x) = v(\exp x)$  would satisfy  $v(\sigma x) = qv(x) = v(qx)$ , using the  $\mathbb{Q}$ -linearity of the additive map  $v$ . Thus  $\sigma(x) - qx$  would be finite for every  $x$ . Choose a positive infinite  $X$ , and write

$$\sigma(X) = qX + e, \quad \sigma(X^2) = qX^2 + f$$

with  $e, f$  finite. Multiplicativity yields

$$q^2 - q + \frac{2qe}{X} + \frac{e^2 - f}{X^2} = 0.$$

The last two terms are infinitesimal. Taking standard part gives  $q^2 - q = 0$ , contradicting  $q > 0, q \neq 1$ .  $\square$

Thus even the very simple field automorphism  $S_2$  cannot be repaired by composing with a value-fixing automorphism to obtain an exponential lift of its value action.

### 11.5 A global surcomplex operation that remembers real exponential

There is no need to postulate a total complex logarithm. The operation

$$L : \mathbb{K}^\times \rightarrow \mathbf{No} \subset \mathbb{K}, \quad L(z) = \log |z| \tag{11.3}$$

is already globally defined, using only the usual real surreal logarithm. Say that  $\alpha$  preserves  $L$  when  $\alpha(L(z)) = L(\alpha(z))$  for every nonzero  $z$ .

**Theorem 11.4** (Logarithmic-modulus automorphisms). *For  $\mathbb{K} = \mathbf{No}(i)$ ,*

$$\text{Aut}_L(\mathbb{K}) \simeq \text{Aut}_{\text{exp}}(\mathbf{No}) \times C_2. \tag{11.4}$$

*Its action on the natural value group has kernel exactly  $\{\text{id}, c\}$ . Marking  $i$  makes that action faithful.*

*Proof.* The range of  $L$  is all of  $\mathbf{No}$ , because  $L(\exp x) = x$ . A commuting automorphism and its inverse therefore preserve  $\mathbf{No}$ . Write it as  $\widehat{\sigma}c^e$ . For  $r > 0$  real-surreal,

$$\sigma(\log r) = \alpha(L(r)) = L(\sigma r) = \log(\sigma r).$$

Thus  $\sigma$  commutes with logarithm and hence with exponential. Conversely, if  $\sigma$  is exponential, the norm equivariance (2.4) proves that both lifts commute with  $L$ . The kernel statement follows from theorem 11.2 and the fact that conjugation fixes every valuation value.  $\square$

Theorem 11.4 is another result already present in [10]. Notice that  $L$  is required to transform *equivariantly*:  $L(\alpha z) = \alpha(Lz)$ . Demanding instead  $L(\alpha z) = Lz$  pointwise would impose a stronger condition.

## 11.6 Local exponentials and derivations are different structures

For  $\varepsilon \in \mathfrak{m}$ , the Hahn series

$$\text{Exp}(\varepsilon) = \sum_{n \geq 0} \frac{\varepsilon^n}{n!}, \quad \text{Log}(1 + \varepsilon) = \sum_{n \geq 1} \frac{(-1)^{n+1} \varepsilon^n}{n}$$

are well-defined by positive-support summability. Any strongly  $\mathbb{C}$ -linear element of  $G_v$  commutes with these operations, simply by applying strong summation and the field laws. This includes the phase twists, the exponent dilations, and the constant-fixing flows above. For finite  $z = a + \varepsilon$ , such an automorphism also respects the finite exponential  $e^a \text{Exp}(\varepsilon)$ . None of this implies compatibility with the total real surreal exponential on infinite arguments, as  $S_2$  and proposition 11.3 demonstrate.

The Berarducci–Mantova construction equips  $\mathbf{No}$  with a nonzero derivation compatible with substantial exponential and transseries structure [7]. Fix one such derivation  $\partial$  and extend it to  $\mathbb{K}$  by

$$\partial(x + iy) = \partial x + i\partial y.$$

For real-axis-preserving automorphisms, a direct coordinate calculation gives

$$\{\alpha \in G_{\mathbf{No}} : \alpha\partial = \partial\alpha\} \simeq \text{Aut}(\mathbf{No}, \partial) \times C_2.$$

This is a reduction, not a classification of the differential group or of all differential automorphisms without a marked real axis. In particular, the scalar derivation  $\partial$ , formal differentiation of a power series, and the fine derivative of a function are three distinct notions. The zero-derivative automorphism in theorem 8.2 concerns the third.

## 12 Consequences for algebra, geometry, and symbolic computation

Automorphisms transport identities, but a transported identity must include all its coefficients and all its chosen structures. This elementary rule is especially useful when applying surcomplex theory.

### 12.1 Polynomial and finite-dimensional invariants

If  $p(X) = \sum_{k=0}^n a_k X^k$  and  $\alpha$  is a field automorphism, put  $p^\alpha(X) = \sum_{k=0}^n \alpha(a_k) X^k$ . Then

$$\alpha(p(z)) = p^\alpha(\alpha z).$$

Roots, root multiplicities, common-root relations, resultants, discriminants, and finite matrix determinants therefore transform algebraically. When all coefficients are fixed, the root multiset is permuted. Without coefficient fixation, it is the roots of the *transformed* polynomial that are obtained.

For  $\alpha \in G_v$ , root separations satisfy

$$v(\alpha r_j - \alpha r_k) = \tau_\alpha(v(r_j - r_k)).$$

Thus finite valuation-cluster trees retain their combinatorial incidence, although the labels can be reindexed. For  $\alpha \in \mathcal{U}$ , every nonzero separation even retains its leading term. These statements follow from additivity and the valuation laws; they require no compactness or ordinary analytic convergence.

For  $\alpha \in G_{\mathbf{No}}$ , Hermitian algebra is also transported, because  $\alpha$  commutes with conjugation. The identity  $A^* = \overline{A}^T$  transforms to  $(\alpha A)^* = \alpha(A^*)$ , and positive real scalars remain positive. Under a phase twist not preserving the axis, the same conclusion holds only after conjugation is also transported to  $j_\theta$ . Keeping the original conjugation while changing the scalars is not an invariant operation.

## 12.2 Automorphism tests as checks on a proposed canonical definition

Suppose a construction is claimed to depend only on the pure field  $(\mathbb{K}, +, \cdot)$  and on a specified set of scalar parameters. It must commute with every automorphism fixing those parameters. Proposition 9.3 then immediately rules out pure-field definitions of the distinguished real axis and finite ring. A purported canonical modulus on the bare field would similarly determine extra real-form information and cannot be the chosen modulus without additional structure.

This does not make those constructions invalid. It identifies which data must be declared: conjugation or a real form for Hermitian geometry, a valuation for scales, monomials and a summation structure for normal-form algorithms, and an exponential for logarithmic growth. An automorphism counterexample is often the quickest way to detect an omitted hypothesis.

## 12.3 What a symbolic implementation can realistically support

A symbolic system can implement (4.1) exactly on a declared Hahn workspace once it can evaluate the coefficient automorphism, exponent map, and character, and certify that the exponent map is an ordered additive bijection. For finite supports, the arithmetic checks reduce to finite convolution. For symbolic infinite families, well-order and finite-incidence certificates remain necessary.

The maps  $S_\alpha$ ,  $D_\lambda$ , and  $P_\theta$  are particularly transparent on normal-form expressions with accessible exponent coefficients. The fixed-shift flow can be represented by its operator formula or by closed expressions such as  $t \mapsto t/(1 - st)$ . The coefficient-motion example requires an abstract derivation on real constants and should not be advertised as an algorithm on arbitrary numerical real inputs. Likewise, the back-and-forth existence theorems are not executable algorithms on unrestricted surcomplex names.

## 13 Relation to the Surreal repository and a formalization plan

This section is the source’s comparison, as of its pin. It was checked for this report and is accurate at the pin; Section 1.5 adds the relations it did not draw, and Appendix C.2 records what has changed since.

The inspection in this article uses the repository revision

dcf86662b574988e75d08d515d3a2a5ba7bbc0d7.

The root documentation, the report catalogue, the exponential-rigidity manuscript and its audit, the normal-form bridge note, and the two specific Lean modules below were read through the GitHub connector. This is a selected-source inspection, not a claim to have searched every file or to have rebuilt the project.

The inspected documentation distinguishes research arguments from checked Lean coverage. In particular, the current bridge note reports an ordered field equivalence for small normal forms, transfer of real closedness, actual surcomplex algebraic closedness, and valuation/standard-part preservation [11]. These are *repository-reported* implementation facts. This article supplies no independent build result for them.

### 13.1 Existing interfaces actually inspected

Surreal/Algebra/Complexify.lean defines the quadratic algebra  $F[i]$ , its imaginary unit, coordinate multiplication, conjugation, normSq, and the identity mul\_conj. Its field construction uses an ordered coefficient field and explicitly does not, by itself, assert algebraic closedness [12]. These interfaces are sufficient foundations for a generic development of theorems 2.2 to 2.4.

`Surreal/Algebra/FineDerivative.lean` defines `FineHasDerivAt` by punctured-neighborhood difference quotients and contains uniqueness, continuity, algebraic differentiation rules, and the polynomial derivative theorem [13]. It is an appropriate interface for theorem 8.1; the exponent-dilation example additionally requires a proved normal-form automorphism and its exact valuation formula.

The documentation’s statements about smallness must retain their universe indices. A theorem that every lower-universe subset is closed and discrete does not say that every subset available in a larger universe is discrete. The class theory in this article and a universe-relative Lean realization are compatible approaches, but they are not textually interchangeable.

### 13.2 Suggested implementation order

A useful order is to separate finite algebra from normal-form and global existence arguments.

1. **Generic real-form algebra.** Construct the lift of a base-field ring equivalence to  $F[i]$ . Prove uniqueness from its restriction and its value on  $i$ , and identify the conjugation centralizer. Add exact norm rigidity and the Cayley-circle criterion. These proofs need no Hahn theory.
2. **Explicit Hahn automorphisms.** Package coefficient maps, ordered additive exponent equivalences, and characters in one construction. Prove the composition law (4.2) and valuation formula before introducing the phase action. Keep the nested coefficient extraction  $\ell$  on surreal *exponents* distinct from coefficient extraction on a surcomplex input.
3. **Support-certified flows.** Reuse a summable-family interface and prove the fixed-shift family is jointly summable. Establish inverse formulas for theorems 5.1 and 5.2 through coefficientwise finite identities. Merely entering a formal expression  $\exp(D)$  is not a proof of an automorphism.
4. **Topology and derivatives.** Prove the induced homeomorphism from a valuation reindexing, the nonzero-derivative rigidity theorem for a generic topological field, and finally  $S'_a \equiv 0$  for  $a > 1$  using (8.1). No mean-value theorem is needed or available for that last conclusion.
5. **Global foundations separately.** Treat class back-and-forth, real-form existence, and collection-level group factorizations in an explicit foundational layer or in bounded universes with the appropriate hypotheses. Do not silently replace a set-stage extension theorem by an unrestricted type-level statement about every model.

The names in this implementation order describe proposed work; they are not assertions that corresponding new modules already exist. In particular, no new Lean code or Lean-checking claim is included in the present package.

## 14 What remains to be classified

The central stabilizer question has a clean answer, and the valued subgroup has an exact kernel–quotient decomposition. Neither result gives a fully explicit description of the unrestricted automorphism collection. The following are useful research targets, with their status kept distinct.

**The image of exponential automorphisms on the value group.** Faithfulness (11.2) determines the kernel, not the image. An intrinsic necessary-and-sufficient criterion on an ordered additive  $\tau : \mathbf{No} \rightarrow \mathbf{No}$  for it to come from an exponential automorphism would be much stronger. Positive rational dilations already show that arbitrary Hahn lifting does not solve the problem. Any such criterion must use the interaction between the value group and real exponential growth, not just ordered-group structure. (This is Question 13.1, `q: image`, of the collection’s rigidity report, where it is also open.)

**Omega-map compatibility.** The nontrivial automorphisms preserving the omega-map, and extensions of order automorphisms of its generalized epsilon fixed points, are posed in Questions 5.6 and 5.7 of the inspected version of [5]. They are not resolved here. Exact value rigidity only describes one kernel and must not be mistaken for a solution of those broader extension problems.

**Effective descriptions inside the leading-term kernel.** The abstract support-and-invertibility conditions for strong Hahn substitutions are meaningful, but finite, compositional certificates for larger symbolic families would be useful for computation and formalization. The fixed-shift constructions are a tractable subclass, not an exhaustive normal form for  $\mathcal{U}$  or its strong part. This is a proposed computational research direction, not a claim that the general Hahn literature lacks abstract descriptions.

**Real-form and continuity classifications.** The phase family and the cofinality example show that real forms have both embedding freedom and isomorphism-type freedom. A detailed classification with additional valuation, summation, or simplicity conditions is a separate problem. Similarly, this article proves that  $G_v$  acts continuously but does not identify every fine-continuous pure-field automorphism. These are boundaries of the present account; they are not asserted to be newly formulated or historically unresolved in every related setting.

## 15 Source audit, proof status, and conclusions

### 15.1 Version-sensitive points

The version-specific literature anchor is Kaplan–Krapp–Serra, arXiv:2509.22374v3, submitted 23 April 2026, whose PDF is dated 27 April 2026. Its displayed formulas were checked against the PDF, not only the HTML rendering. Two local cautions affect example construction. The shift  $g - 1$  printed in Example 4.3 is not contracting for the convention  $v = \min \text{supp}$ ; our construction uses a positive shift. The series  $\sum_n \omega^{-1/n}$  printed in Example 4.4 does not have decreasing Conway support; (7.1) uses a valid support instead. Our nonstrong example does not claim the stronger global leading-term property asserted for that source’s construction.

The semidirect actions in this article are specified by actual composition formulas, especially (4.2) and (6.8). A display listing factors should not be read as a componentwise direct product when exponent reindexing acts on coefficient weights.

The repository’s exponential-rigidity report is explicitly prior material at the pinned revision. Its appended source audit records an earlier catalogue/tree mismatch, illustrating why absence from a catalogue or failure of code search is not evidence of absence from the repository. The present article neither relies on that earlier absence claim nor makes an exhaustive nonduplication claim.

## 15.2 Evidence categories

| Category                | Treatment in this article                                                                                                                                                 |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classical prerequisites | Real closedness and normal forms of <b>No</b> ; Newton–Puiseux; finite-group fixed-field theory and Artin–Schreier are imported with references.                          |
| Direct proofs           | Real-axis, circle, and norm stabilizers; explicit Hahn maps and flows; valued factorization; topology and derivative results; class extension and real-form consequences. |
| Repository prior work   | Exponential displacement faithfulness, rational non-lifting, and logarithmic-modulus reduction are attributed, with proofs included for their role here.                  |
| Implementation evidence | Selected source code and documentation were read; no independent repository build and no new Lean certification.                                                          |
| Historical novelty      | Not certified. Derived statements and useful examples are not automatically claimed to be first discoveries.                                                              |

## 15.3 Non-claims

The source states its limitations where they arise, and each is kept in place. The list collects them, with the place where each stands. Items 1–28 are the source’s; items 29–33 were added when the report joined the collection.

1. No complete classification of the automorphisms of  $\mathbb{K}$ , and in particular none of the unrestricted ones, is claimed (title page, abstract, Section 14; the delivered README).
2. Historical priority is not claimed or certified; derived statements and useful examples are not automatically claimed to be first discoveries, and no exhaustive nonduplication claim is made (title page, Section 15; the delivered README).
3. No new Lean verification, no Lean code, and no independent build of the repository are claimed; the implementation order describes proposed work, not existing modules (title page, Section 13, Section 15).
4. The repository comparison is a selected-source inspection, not a search of every file; the bridge-note facts are repository-reported, not independently rebuilt (Section 13).
5. Exponential faithfulness, the rational non-lifting proposition and the logarithmic-modulus reduction are prior work of the repository, not results of this report; the settlement of Question 5.4 of [5] is by the repository’s proof and “is not a new result of this article” (abstract, Sections 1 and 11, Section 15).
6. Faithfulness does not say that the exponential automorphism group is trivial (Section 11).
7. The article does not rely on the earlier absence claim in the rigidity report’s audit (Section 15).
8.  $\text{Aut}(\mathbb{K})$  and the other groups are definable collections of class maps, not class objects of NBG; a group isomorphism is a unique factorization statement about individual automorphisms (Section 1).
9. Global homogeneity and the topology of the full field are not to be transferred to an arbitrary fixed set-sized workspace (Section 1).
10. The four-layer decomposition (6.2) is not a decomposition of all field automorphisms of  $\mathbb{K}$ , and its factor  $\mathcal{U}$  is not reduced to a finite list of coefficients (Section 6).
11. An arbitrary homomorphism  $g \mapsto u_g$  into  $1 + \mathfrak{m}$  does not by itself give a summable or surjective map; the fixed-shift flows illustrate the automorphism–derivation connection without asserting that every formal operator exponential is summable (Section 5).

12. The Taylor motions  $\Phi_d$  are generally noncomputable from numerical names of real coefficients and are not to be advertised as algorithms; the back-and-forth existence theorems are not executable algorithms and give no canonical parametrization of the class automorphisms (Sections 5, 9 and 12).
13. The maps of  $G_{\mathbf{N}_0}$  are semilinearly compatible with the modulus; calling them isometries would be misleading (Section 2).
14. Proposition 7.1 is an inclusion in the group of fine-continuous automorphisms, not a classification of it, and no converse is inferred from continuity; not every fine-continuous pure-field automorphism is identified (Sections 7 and 14).
15. The non-strong witness of Section 7 is not claimed to fix every value or every leading term, nor to have the global leading-term property asserted for the construction in [5]; the non-density conclusion concerns the pointwise-equality topology on automorphisms only (Sections 7 and 15).
16. The choice of topology is part of each topological theorem; the discreteness of set-sized subsets in the full fine topology says nothing about the intrinsic valuation topology of a fixed Hahn workspace (Section 7).
17.  $S_2$  is not asserted to be representable by a locally convergent power series; “everywhere differentiable” is not a synonym for analytic rigidity, and a zero derivative does not imply local constancy (Section 8).
18. The scalar Berarducci–Mantova derivation, formal differentiation of a power series and the fine derivative are three distinct notions; the differential reduction is not a classification of the differential group or of differential automorphisms without a marked real axis (Section 11).
19. The transcendence-basis description is not a classification, even for a set-sized algebraically closed field (Section 9).
20. Neither the phase family nor Theorem 10.2 classifies the real closed class fields whose complexification is  $\mathbb{K}$ ; the conjugacy criterion of Appendix B is structural, not an effective decision procedure (Section 10, Appendix B).
21. Simplicity rigidity is rigidity of an enriched structure, not of the real closed field reduct (Section 11).
22. Exact value rigidity for the omega-map needs pointwise value fixation; it describes one kernel and is not a solution of the omega-map extension problems, Questions 5.6 and 5.7 of [5], which are not resolved (Sections 11 and 14).
23. Compatibility with the local exponential and logarithm does not imply compatibility with the total real exponential on infinite arguments (Section 11).
24. The image of exponential automorphisms in the value group is not determined (Section 14).
25. Automorphism tests do not make the constructions they constrain invalid; they identify data that must be declared (Section 12).
26. Smallness statements keep their universe indices; the class theory here and a universe-relative Lean realization are compatible but not textually interchangeable (Section 13).
27. The fixed-shift constructions are a tractable subclass, not an exhaustive normal form for  $\mathcal{U}$ ; the research targets of Section 14 are not asserted to be newly formulated or historically unresolved in every setting, nor is it claimed that the Hahn literature lacks abstract descriptions (Section 14).
28. The delivered README reports symbolic finite checks of the norm product and polarization identities, the Cayley inverse, and the rational-flow composition and inverse identities, and says that such checks are not proofs of infinite summability or of the full surreal theory. No code or record of them was delivered, and nothing in this report rests on them. The delivery redistributes no third-party papers or font files.

29. (*Added.*) The answer to Question 5.4 of [5] rests on the rigidity report's check of the pinned version of that paper. The source's readings of its Examples 4.3 and 4.4 and of its Questions 5.6 and 5.7 are the source's and were not rechecked for this report.
30. (*Added.*) Whether  $\text{Aut}_{\text{exp}}(\mathbf{No})$  is nontrivial, equivalently whether  $\{\text{id}, c\} \subsetneq \text{Aut}_L(\mathbb{K})$ , is not decided; nor is whether  $S_a|_{\mathbf{No}}$  is exponential when  $a$  is not rational (Section 1.4).
31. (*Added; status updated.*) No dedicated `saut`: mappings are supplied. Existing generic Lean proofs cover the ordered displacement and value-fixing exponential rigidity implications, as recorded in Section 1.5; their actual surreal exponential instantiation remains pending. The discreteness proof concerns lower-universe-small subsets of the Lean carrier, not an unqualified proper-class statement.
32. (*Added.*) The relations of Section 1.5 identify results by their content; no theorem is transferred between the full fine topology and a workspace topology, or between the fine derivative and another derivative notion.
33. (*Added.*) Observation (1.2) is an elementary consequence of results printed here; it is not claimed as a result of the source or of the rigidity report.

## 15.4 The main picture

The correct starting point is the strict separation

$$\{\text{id}, c\} = \text{Aut}(\mathbb{K}/\mathbf{No}) \subsetneq G_{\mathbf{No}} \subsetneq G_v \subsetneq \text{Aut}(\mathbb{K}).$$

Preserving the real axis gives  $\text{Aut}(\mathbf{No}) \times C_2$ ; preserving the unit circle gives the same group. Exact preservation of every modulus reduces it to identity and conjugation. Pure-field automorphisms can move the real axis, the finite ring, and even the isomorphism type of a chosen real form under a change of conjugation.

The valuation organizes a large explicit subgroup but does not enforce classical analytic behavior. Phase twists can move real monomials while fixing constants and values; leading-term automorphisms can move ordinary real constants by infinitesimals; fine-continuous maps need not preserve Hahn sums; and exponent dilation can be a differentiable field homeomorphism with derivative zero everywhere. Exponential structure imposes a different and genuinely strong constraint: its value-group action is faithful, even though the underlying valued-field kernel is large.

An automorphism theorem for surcomplex numbers is therefore meaningful only after its preserved structure, support convention, topology, and size framework have been made explicit. With those choices in place, the theory has both exact structural reductions and a rich supply of concrete examples.

## A A quick reference of explicit examples

In the table, “values fixed” means  $v(\alpha z) = v(z)$  for every nonzero  $z$ , not merely preservation up to a value-group automorphism. The term “constants” means the embedded ordinary complex field.

| Map                               | Preserved data                                                          | Nontrivial feature                                                                              |
|-----------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| $c$                               | Real axis pointwise; modulus; values; strong sums                       | $i \mapsto -i$ ; not complex fine-differentiable                                                |
| $S_2$                             | Constants; real axis; conjugation; strong sums                          | $v \mapsto 2v$ ; $\omega \mapsto \omega^2$ ; derivative zero                                    |
| $D_\lambda, \lambda \neq 0$       | Constants; real axis; conjugation; values; strong sums                  | Rescales leading coefficients by $e^{\lambda(g)}$                                               |
| $P_{\pi/2}$                       | Constants; values; strong sums; fine topology                           | $t \mapsto it$ ; moves the real axis and changes modulus                                        |
| $\Phi_d, d(b) = 1$                | Monomials; leading terms; strong sums; real axis when $d$ is real       | $b \mapsto b + t^\delta$ ; ordinary constants need not be fixed                                 |
| $T_1, \lambda = \ell, \delta = 1$ | Constants; real axis; conjugation; leading terms; strong sums           | $t \mapsto t/(1-t)$ ; not a total-real-exponential automorphism                                 |
| Ordered extension of Section 7    | Constants; real axis; conjugation; fine topology                        | Fixes the summands of $a$ but sends $a \mapsto a + t^\omega$                                    |
| Pure-field back-and-forth         | Any prescribed set of parameters, with ordinary $\mathbb{C}$ also fixed | Can send a positive infinite $x$ to $x^{-1}$ or $ix$ ; no canonical normal-form action supplied |

## B A conjugacy criterion for the standard real form

The cofinality obstruction in [theorem 10.2](#) has a useful positive counterpart. Say that an ordered class field has the *set-cut interpolation property* if for every pair of sets  $\mathcal{L}, \mathcal{R}$  with  $\mathcal{L} < \mathcal{R}$ , there exists  $x$  with  $\mathcal{L} < x < \mathcal{R}$ . Empty sides are permitted.

**Proposition B.1.** *An involution  $j$  of  $\mathbb{K}$  is conjugate to  $c$  if and only if its fixed real field has the set-cut interpolation property.*

*Proof.* The property holds in **No** by Conway cuts and is preserved by ordered isomorphisms. Conversely let  $F = \text{Fix}(j)$  have the property. It is a proper class, since  $\mathbb{K} = F(i)$ , and it is real closed by [theorem 10.1](#). Run the proof of [theorem 9.2](#) between  $F$  and **No** rather than within **No**. At every stage, the only special existence condition used to extend the partial ordered-field isomorphism is interpolation of a set-sized cut. It holds on both sides, so back-and-forth gives  $F \simeq \mathbf{No}$ . The real-form conjugacy criterion in [Section 10](#) then gives  $j \sim c$ .  $\square$

This is a structural characterization, not an effective decision procedure for a given class involution. It singles out precisely the conjugacy class of the usual surreal real form among all real forms of the pure field.

## C Provenance and the pinned comparison

### C.1 Provenance of this report

This report is built from *one* manuscript: the archive `surcomplex_automorphisms`, manuscript 07 of batch 20 of the collection, dated 22 September 2026 and pinned to repository commit `dcf86662b574988e`

75d08d515d3a2a5ba7bbc0d7. It was placed in commit 5fe7f8d as a new report rather than merged into surreal/exponential-automorphism-rigidity: it continues that report’s section on **No**(*i*), but with plain field automorphisms instead of exponential ones, and answers no question that report names. There was no second source; nothing was merged and nothing was selected away. Every result, proof, example, question and limitation of the manuscript is printed here.

The delivery consisted of the article source, its PDF and a README; it contained no code and no data. The source was renamed `article.tex` and edited as listed below; the README was rewritten for the collection; the delivered PDF is not shipped and is replaced by a build of this text.

The editorial changes are these.

- (1) Every label carries the prefix `saut:`. The source’s 79 labels are kept, unchanged after the prefix, and seven were added (`saut:sec:conventions`, `saut:sec:position`, `saut:eq:Lchain`, `saut:sec:nonclaims`, `saut:app:conjugacy`, `saut:app:provenance`, `saut:app:pinned`).
- (2) The title page gives the report’s status and provenance, and the abstract gains one sentence placing  $\text{Aut}_L(\mathbb{K})$ .
- (3) Sections 1.4 and 1.5 and the consolidated non-claims of Section 15.3 are new. Short pointers were added after Theorem 6.1, at the start of Section 13, and in the first question of Section 14.
- (4) Six symbols were renamed so that each has one meaning (Section 1.4):  $E_{\rho,\tau} \rightarrow M_{\rho,\tau}$ ,  $E_d \rightarrow \Phi_d$ , the derivation  $D \rightarrow \mathcal{D}$ , the Puiseux field  $P \rightarrow \mathcal{P}$ , the automorphism  $\tau \rightarrow \psi$  of the non-density argument, and cut sides  $(L, R) \rightarrow (\mathcal{L}, \mathcal{R})$ ; the source’s  $\text{Aut}(\mathbb{K}, L)$  is written  $\text{Aut}_L(\mathbb{K})$ .

The placement edit preserved the source’s mathematical statements. It retained the results reproduced from the rigidity report (its Lemma 3.1, the faithfulness of Corollary 4.2, the content of Theorem 5.1 and Corollary 5.2, Theorem 8.4, and Theorem 9.2 with its kernel clause). They are printed here once, with the source’s proofs and attribution, rather than replaced by pointers, because they are part of one manuscript’s argument; Section 1.5 names each counterpart. The letter *E* was freed for the exponential, following the rigidity report, rather than renaming that report’s symbol. The title is the source’s; the directory is named `surcomplex-field-automorphisms` to say that the automorphisms are field automorphisms.

The later proof review clarifies rational cuts and strong coefficient action. It expands the class and support arguments and excludes every derivative value in  $\mathbb{K}$ , including infinite values. The finite-observation non-density proof is extended to all surcomplex field automorphisms. Current generic Lean coverage is separated from the pinned inspection. See `REVIEW.md` for the changes and validation. The added arguments are not attributed to the original delivery.

## C.2 Pinned repository scope

Section 13 is the source’s comparison at its pin `dcf8666`, an ancestor of the placement commit `5fe7f8d` (sixteen commits earlier). Checked against the repository for this report:

- The descriptions of `Surreal/Algebra/Complexify.lean`, `Surreal/Algebra/FineDerivative.lean`, the normal-form bridge note, and the rigidity report with its appended audit correction are accurate at the pin. None of these files, nor the root `README.md`, changed between the pin and the placement; the report catalogue `docs/README.md` changed only by listing reports of another batch and extending other entries.
- The bibliography cites the rigidity report by its article title, *Growth-Scale Rigidity for Surreal and Surcomplex Automorphisms*; the collection’s README for the same directory is headed “Cofinal displacement and the rigidity of exponential automorphisms”. Both name `surreal/`

exponential-automorphism-rigidity.

- The source did not relate its results to `found.thm:discrete` and its Lean proof, to the collection's zero-derivative witnesses, to the character twists of `spec.lem:finitenessindex`, to the Cayley parametrization of the trigonometry report, or to the Lean modules `Surreal/Surcomplex/Basic.lean` and `Surreal/HahnSeries/ComplexNumbers.lean`, all present at the pin. The hidden-negative-directions and bounded-support reports and `Surreal/Algebra/SigmaDerivation.lean` were added after the pin and could not have been seen. Section 1.5 supplies these relations.
- The README of the rigidity report states that nothing there proves a theorem about the automorphism group of **No** that any other report also proves. That was accurate before this report was placed; this report reprints several of its results (Section 1.5).

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